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A multi-objective sustainable load planning model for intermodal transportation networks with a real-life application

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ABSTRACT

Growing importance of intermodal transportation necessitates modeling and solving load planning problems by taking into account various complex decisions simultaneously like transportation mode/service type selection, load allocation, and outsourcing. This paper presents a mixed-integer mathematical programming model for a multi-objective, multi-mode and multi-period sustainable load planning problem by considering import/export load flows to satisfy transport demands of customers and many other related issues. Several multiple objective optimization procedures are utilized in order to handle conflicting objectives simultaneously under crisp and fuzzy decision making environments. A real-life case study is also performed to present application and usefulness of the proposed model.

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1. Introduction

Nowadays, there is a rapid increase in intermodal transportation research in the literature. The intermodal transportation was defined as "transportation of freights in one and same intermodal load unit (container, trailer, swap, box, body, etc.) by using successive modes of transport (railway, maritime, road, etc.) without handling the freights themselves when the transportation mode is altered" (UNECE, 2009). There are some different terminologies, i.e., multimodal, intermodal, co-modal and synchro-modal, for the multi-mode transportation systems in the current literature and global logistics industry. Actually, multimodal transportation is the broadest term and covers all of the others (SteadieSeifi et al., 2014). In detail, multimodal transportation is able to provide more efficient, reliable, flexible and sustainable way of freight transportation. Intermodal transportation and containerization can also support the transport flexibility and give advantage of economies of scale to the logistics companies. Because, more efficient usage of company sources can be yielded resulting from the integration of different transportation modes, improvements in handling speed and throughput (Rodrigue and Slack, 2013). In fact, popularity of the intermodal transportation which is the most widely utilized multimodal term, takes root from the problems of road congestion, environmental responsibilities and traffic safety. In other words, intermodal freight transportation planning problem has become a seriously studied research topic since its ability to increase traffic safety, customer satisfaction and decrease transportation costs, road congestion and environmental pollution. The intermodal freight transportation combines at least two different transportation modes in a single transport chain. In general, a transport chain mainly consists of three components namely pre-haulage, long-haulage and end-haulage which correspond to pick-up, door-to-door transfer and delivery services, respectively. The pre and end-haulages correspond to the trucking part of a transport

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chain and also named as drayage activities. These initial and final journey activities are mostly performed by road freight transportation due to the short travel distances. Drayage costs constitute the majority portion of the overall transportation costs even though the relatively short distance of truck movements. For this reason, drayage activities may have a significant impact on the profitability of intermodal transportation systems (Macharis and Bontekoning, 2004). In addition to these pre and end-haulage activities, transfer of the load units between any two terminals, i.e., seaports, railway stations and hubs, is called as long-haulage. In contrast to the large proportion of trucking for inland transportation, many logistics companies tend to prefer railway, airline, maritime transportation services and their combinations for long-haul freight transportation in Europe (UIC, 2014; SteadieSeifi et al., 2014). A typical intermodal transport chain including both short and long haulages can be portrayed as in Fig. 1. Although there are several studies on the logistics network design, transportation mode and service type selection, hub/terminal location and vehicle routing problems, there are still a lack of studies in the literature for freight planning problems of intermodal transportation networks. Actually, most of the previous studies have focused on single mode freight transportation planning problems from the perspective of logistics service providers or production companies which undertake the logistics activities by themselves. In other words, there are a few papers on intermodal freight transportation planning problems when compared to the unimodal transportation systems. On the other hand, planning problems in intermodal freight transportation may become much more complex than the traditional unimodal transportation systems. Therefore, they may require development of more comprehensive mathematical models, advanced Operational Research (OR) and Artificial Intelligence (AI) based solution methodologies. Fortunately, multi-mode load planning or intermodal freight transportation planning problems have become attractive research areas in recent years. Because, there has been an increasing interest to multimodal transportation systems by both researchers and practitioners in the global logistics sector all over the world. Besides, environmental aspect of the intermodal transportation systems was highlighted as current research trends by Caris et al. (2013) since it plays an important role in policy making. On top of all these, development of decision support systems based on advanced optimization models and innovative applications of AI techniques for intermodal freight transportation planning problems was given as a future research direction by Macharis and Bontekoning (2004).

Based on these motivations, this paper presents a new mixed-integer programming model for a multi-objective, multi-mode and multi-period sustainable load planning problem. In other words, a sustainable intermodal freight transportation planning model is developed based on network flow modeling for tactical planning activities. The main contributions of this research can be listed as follows: A novel mathematical programming model is proposed for a real-life intermodal freight transportation planning problem of a large-scaled international logistics company in Turkey. The proposed model is able to handle various critical decisions concurrently such as transportation mode/service type selection, outsourcing, periodic load allocation, consolidation at the transshipment seaports/rail stations, usage frequency of block train services and Ro-Ro vessels. Both of the import and export load flows are considered by the proposed model in order to satisfy the bi-directional transport demands of the customers. Consolidation operations or exchange of load units at the transshipment seaports/railway stations are also incorporated into the proposed model. For the maritime logistics operations, different transportation services by company-owned Ro-Ro vessels and other logistics service providers' Ro-Ro vessels are taken into account while generating the load plans. In addition to the railway transportation services by block trains with fixed time schedule, alternative public train services with flexible time schedule are also considered within the proposed model. As well as to minimize the overall transport costs, maximal customer satisfaction is targeted by minimizing the total road, marine and railway transit times. Furthermore, environmental conscious load plans are generated by minimizing the amounts of total CO_2 emitted by the different transportation modes. Therefore, sustainable load plans can be produced by making

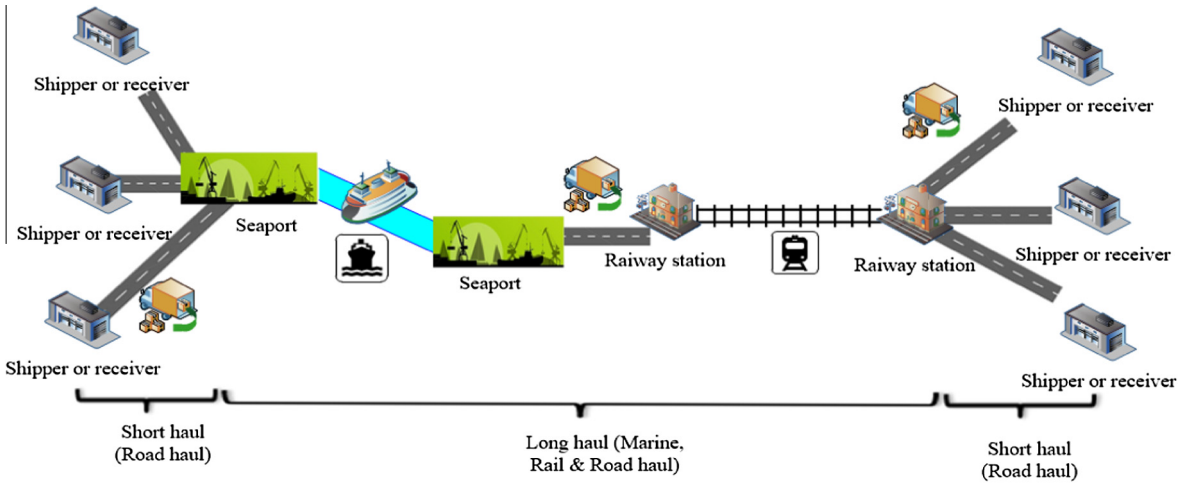


Fig. 1. A simple depiction of road-marine-rail intermodal freight transport.

use of the proposed model in terms of cost savings, high level of customer service and environmental gains. In order to present applicable compromise solutions to the company experts, different multi-objective optimization techniques, i.e., compromise programming, fuzzy goal programming with different importance and priorities, interactive fuzzy goal programming and weighted additive fuzzy goal programming, are applied under both crisp and uncertain environments. The extensive computational analysis have shown that increase in usage rate of intermodal transportation may provide significant benefits for all of the company objectives in terms of cost, time and environment. Furthermore, it can also be reported that compromise programming technique with crisp goals may produce superior results over the other multi-objective optimization approaches in terms of total cost and environmental objectives. On the other hand, the best results for transit time objective can be found by using interactive fuzzy goal programming approach under uncertainty of aspiration levels. Finally, scenario analysis have also demonstrated that marine transport cost of other logistics service providers' Ro-Ro vessels, upper bounds on number of block train services, lower bounds on utilization rates of the company-owned Ro-Ro vessels and portion of road transportation in outsourcing option can be seen as the most influential factors on the company's goals.

The remainder of this paper is organized as follows. In Section 2, a comprehensive literature review on planning problems in intermodal transportation with OR and AI based solution methodologies is presented. In Section 3, details of the problem with logistics network representation, modeling assumptions, nomenclature and mathematical model formulation are described. Section 4 presents a real-life application for model verification and validation. This section also contains the results of deterministic single objective and fuzzy multi-objective optimization models. Scenario analysis is also carried out for providing managerial insights at the end of Section 4. Concluding remarks and future research directions are given in Section 5.

2. Related literature

An overview of the recent articles which were studied the multi-mode load planning, intermodal route selection and service network design problems with relevant applications in multimodal transportation environments can be summarized as follows: A review and classification of OR models and techniques implemented in intermodal transportation research area were presented by Macharis and Bontekoning (2004). They considered the goals/specific decisions of different operators and length of the planning horizon. In other words, they categorized the existing studies according to type of decision makers, i.e., drayage operator, terminal operator, network operator and intermodal operator, and length of the planning horizon (strategic, tactical and operational level problems). These operators with different fields of interest can be summarized as depicted in Fig. 2. The relevant classification matrix for intermodal operations problems is also given in Table 1. Then, Caris et al. (2008) extended that review and presented a more comprehensive literature survey on planning problems in

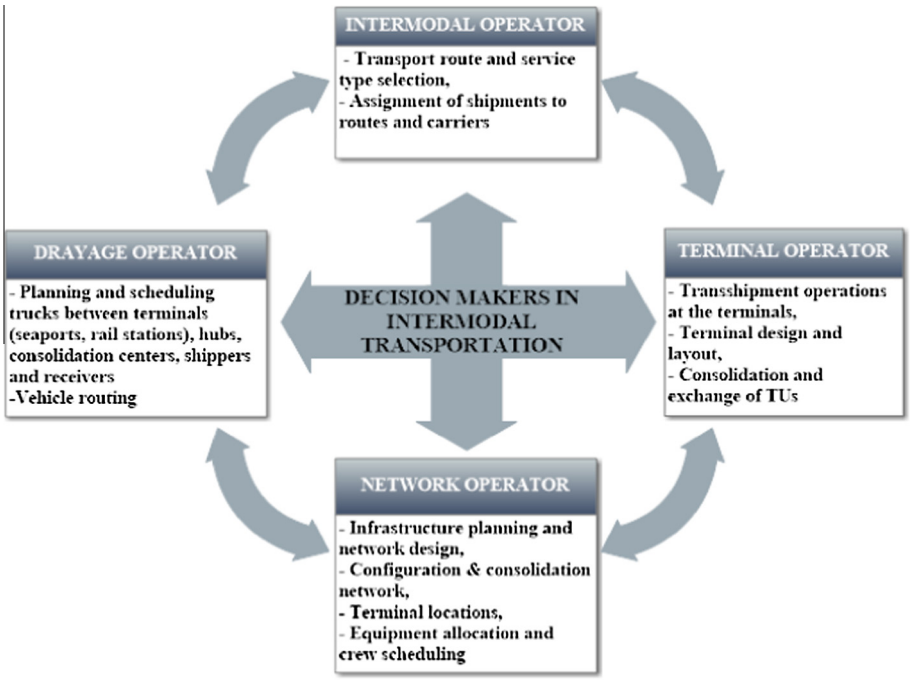


Fig. 2. Multiple decision makers and their fields of interest in intermodal transportation.

Table 1
Classification of operations problems in intermodal transportation networks (Macharis and Bontekoning, 2004).

Decision maker	Field of interest	Time horizon		
		Strategic	Tactical	Operational
Drayage operator	–Planning & scheduling of trucks between the terminals, shippers & receivers	– Co-operation between drayage companies (central planning), – Truck, trailer and chassis fleet size	– Assignment of shippers and receiver locations to terminals, – Pricing strategies	– Repositioning of trailer chassis and load units, – Scheduling of truck trips, – Vehicle routing
Terminal operator	–Transshipment operations	– Terminal design and layout, – Consolidation and exchange of load units	– Capacity levels of equipment & labor, – Redesign of operational routines and layout structures	– Reschedule capacity based on real-time data, – Allocation of capacity to jobs & job scheduling, – Resource allocation
Network operator	–Infrastructure planning & network design	– Determining the terminal locations, – Infrastructure configuration, – Network design	– Configuration consolidation network, – Determine service types, – Determine production model type, – Allocation of equipment to routes, – Optimal price and level of service	– Scheduling of services, – Load order of trains and barges, – Redistribution of railcars, trailers, barges and load units – Assignment of trailers/containers to flatcars, – Crew scheduling
Intermodal operator	– Route selection for shipments	N.A.	N.A.	– Route and service selection, – Assignment of shipments to routes and carriers, – Routing and repositioning

intermodal transportation. It was also emphasized by Caris et al. (2008) that there is a scarcity of researches which cover various time horizons and integrate decisions of more than one decision maker. In another literature survey, Caris et al. (2013) classified the intermodal transportation research field as followings: policy support, terminal design, intermodal service network design, intermodal routing, drayage operations and information/communication technology innovations. An integer programming model which purposes to minimize the total cost of the empty container movements was developed by Choong et al. (2002) to examine the impact of planning horizon length on the mode selection and empty container management in intermodal transportation. A binary integer program based on a multi-commodity fixed-charge network design model was presented by Arnold et al. (2004) for multimodal terminal location problem in a rail/road intermodal transportation system. A heuristic procedure depending upon the shortest path algorithms was also used to solve large scaled problems within a reasonable computational time. The intermodal routing problem which minimizes cost and travel time was formulated by Chang (2008) as a multi-objective, multimodal, multi-commodity network flow problem. The time windows and non-convex piecewise linear cost functions were also taken into account. Since the NP-hard nature of the problem, a heuristic solution approach based on Lagrangian relaxation and decomposition techniques was proposed to separate the original problem into smaller sub-problems.

A collaborative planning approach was developed by Puettmann and Stadtler (2010) for decentralized planning of an intermodal transport chain including three autonomous service providers. These service providers consist of an intermodal operator and two carriers who undertake the drayage activities in the regions of shipper and receiver. In detail, they proposed a coordination scheme in which decisions of these service providers are made by an iterative proposal generation mechanism depending on three different mathematical programs. An integer linear programming formulation was presented by Bauer et al. (2010) for an intermodal scheduled service network design problem. The fleet management issues were also included under a central control assumption. They focused on the energy consumption by all of the intermodal transport activities to minimize greenhouse gas emissions. Verma and Verter (2010) proposed a bi-objective analytical framework to specify the most appropriate freight shipment plan for regular and hazardous materials in a rail/truck intermodal transportation network. The truck route and train service selection decisions were driven by customer specified lead times while optimizing the transport cost and public transport risk. An intermodal network optimization model was developed by Yang et al. (2011) in order to achieve the best routing plan from China to Indian Ocean. Moreover, an integrated cost and goal programming models were utilized so as to handle some conflicting objectives, i.e., minimize cost, transit time and transit time variability, in an aggregated manner.

A decision support framework which contains three core models was developed by Macharis et al. (2011) for economic and environmental policy making in intermodal transportation. This framework involved a multimodal freight transportation model for optimal location of new terminals and a discrete event simulation model for analyzing the potential market area. The final model was used for investigating the influence of new intermodal barge terminal on the network

characteristics such as average waiting times at locks, and turnaround times of terminals. A mixed-integer programming model was proposed by Bierwirth et al. (2012) for tactical planning problem of an intermodal transportation network in Eastern Europe. Their model incorporated the service type selection, mode choice, intermodal terminal selection, freight consolidation for rail transport services and company's production plan. Meisel et al. (2013) extended the model of Bierwirth et al. (2012) by adding multiple type of commodity, production set-up, master production schedule, inventory management, multiple time periods and multi-stage rail transportation services. They utilized both exact and approximate solution approaches such as Branch-and-Cut method and two different decomposition-based heuristics, respectively. Finally, ε-constraint method was employed for handling multiple objectives simultaneously and generation of Pareto-optimal solutions. Three different integer programming models which targeted maximization of the train utilization and minimization of the transportation cost were developed by Bruns and Knust (2012) for load planning problem of trains at an intermodal container terminal. In contrast to the available literature, additional weight restrictions were also taken into account by these models. Besides, the length constraints were also considered while setting of all wagons of a train and assigning the load units to the slots on the wagons. Cho et al. (2012) formulated a multi-objective multimodal routing problem with time windows based on a weight constrained shortest path model. Then, Label setting procedure was applied to obtain Pareto optimal solutions by taking into account the trade-off between transportation cost and time. A bi-criteria optimization model with transport cost and risk objectives was developed by Verma et al. (2012) for a rail-truck intermodal transportation routing problem of hazardous materials. For solution of the mathematical model and derivation of non-dominated solutions, a Tabu search based solution approach was proposed and compared with a constructive heuristic.

Assadipour et al. (2015) extended the study of Verma et al. (2012) by incorporating the congestion at the terminals and terminal equipment capacity decisions into a bi-objective non-linear mixed-integer programming model. A multi-objective genetic algorithm based solution methodology with a non-binary solution representation was also developed due to the non-linear terms into the risk objective and some constraint functions. A multi-objective chance constrained programming model with stochastic transport times was presented by Kalinina et al. (2013) for real time matching of freights and intermodal transport alternatives. They utilized ε-constraint method to generate Pareto optimal solutions while minimizing the cost, transport time and CO_2 emission objectives jointly. A hybrid approach which combines OR and AI techniques was introduced by Garcia et al. (2013) for a container-based specific intermodal transportation problem of a Spanish logistics company. Firstly, three different assignment problems for containers to trucks, trucks and containers to transport routes and trucks to intermodal nodes were solved by both linear programming and a greedy search strategy. Subsequently, automated planning was used to achieve the best transportation mode or path selection for different pick-up and delivery requests of each service. An integrated intermodal freight transport system which includes short-term traffic flow forecasting analysis and an arc flow based mixed-integer programming model was proposed by Bhattacharya et al. (2014) for a rail/road intermodal network optimization. In order to achieve the traffic congestion degree along with different transport routes, a data mining process which considers both space and time dimensions of vehicle data was applied. SteadieSeifi et al. (2014) presented an overview of multimodal transportation literature which involves recent theoretical developments and practical efforts. They also emphasized that there are relatively few papers on operational intermodal planning problems over the strategic and tactical level problems. Additionally, it was mentioned that there is no paper dealt with the dynamic positioning/repositioning of multi-asset resources in intermodal transportation systems. The efficiency of an intermodal transportation system composed of pipeline, barge, rail and truck was evaluated by Kazemi and Szmerekovsky (2015) when designing a multi-echelon, multi-product and multi-mode downstream petroleum supply chain network. Resat and Turkay (2015) developed a novel bi-objective mixed-integer linear programming model for an intermodal transportation network design and operation problem. The traffic congestion on routes was also taken into consideration. In order to produce a set of Pareto efficient solutions for the conflicting transport cost and delivery time objectives, augmented ε-constraint method was utilized. Demir et al. (2015) formulated a multi-objective probabilistic mixed-integer linear program for a green intermodal service network design problem including routing plans. They also dealt with the uncertainty related to travel times and as well as transport demands via chance-constrained stochastic programming approach. The sample average approximation algorithm was implemented in order to provide robust transportation plans with respect to transport cost, service/travel time and greenhouse gas targets. Li et al. (2015) developed an intermodal freight transport network model and multi-level freight transport planning approach for a container flow control problem. The modality changes at the intermodal terminals, limitations on the physical logistics infrastructure, time-dependent transport times, scheduling of trains and barges were also incorporated. In order to cope with the dynamic transport demands and traffic conditions, a receding horizon control approach was proposed and different scenarios were analyzed by performing various simulation experiments. An intermodal logistics network design model with a hybrid network topology was developed by Ghane-Ezabadi and Vergara (2016) to solve integrated terminal/hub location, transportation mode and route selection problems. A path-based composite variable formulation of the problem was also carried out. In order to obtain exact solutions of the large-scaled problem instances within a reasonable computation time, a two-stage optimization approach was utilized by decomposing the original problem into master and multiple sub-problems. Febbraro et al. (2016) introduced a new agent-based modeling framework for short and medium term planning problems in intermodal freight transportation systems. Actually, they proposed a decentralized optimization procedure for cooperative planning of multi-actor systems in order to achieve minimum cost and maximal reliability in intermodal transport chains. The system dynamics were modeled by means of discrete event simulation and freight delivery plans were generated on a rolling horizon basis.

Most of these former researches on intermodal freight transportation systems have concentrated on just single cost or profit objectives and presented hypothetical case studies instead of real-life applications. There are a few studies focusing on the customer satisfaction and environmental pollution resulting from the usage of different transportation modes. Thus, multi-objective sustainable intermodal freight transportation planning models which consider the trade-off between cost, transit time and environmental issues may find more research merit. Moreover, most of the available studies frequently considered only one directional load flows from sources to destinations. Whereas both importation and exportation of loads should be taken into account by the multi-mode load planning models while satisfying the bi-directional transport demands of the customers. This issue is so vital for balancing the import and export freight flows throughout the intermodal logistics network. Another important issue is the transshipment operations at the hubs and terminals. However, load exchanges at the transshipment seaports/railway stations are not generally taken into account in most of the previous models. There is still a scarcity of multi-mode load planning models which incorporate various essential decisions simultaneously such as transportation mode/service type selection, periodic load allocation, outsourcing, marine/rail transshipments, frequency of maritime and railway trips, which makes the problem much more complex but more realistic. Finally, all of the reviewed articles dealt with the intermodal freight transportation planning can be summarized and classified as in Table 2 with respect to the different problem features and modeling characteristics.

3. Problem setting and mathematical model proposal

In this section, a mixed integer programming formulation of a multi-objective, multi-mode and multi-period load planning problem is presented by considering both import and export transportation operations simultaneously. The proposed model has a dynamic nature since it takes into account the variability embedded into the import/export freight demands and periodic trailer/container movements. According to the some criteria, i.e., transit time, tonnage limitations and product characteristics, etc., an international logistics company provides various transportation models and services to its customers including direct road freight transport, combined marine and road transport and intermodal transportation. In direct road freight transport, the loads are transferred from the origin seaports in Turkey to the consolidation centers of the company via highways and delivered again from these centers to the final destinations in the order countries. Similarly, importation begins with the transfer of loads from all over Europe to the consolidation centers via highways for further transportation and then delivered to the relevant seaports in Turkey. The combined marine-road transport integrates two different transportation modes namely marine and road transports and serves as an alternative transportation service for particularly high tonnage products. Because, some tonnage limitations may be imposed in European highways. The main difference between this transportation service and intermodal transportation is the usage of railway transportation operations. In combined marine-road transport, the loads are first transferred from the origin seaports in Turkey to the seaports in Europe by company-owned or other logistics service providers' Ro-Ro vessels and then delivered from these ports to the relevant destinations via road freight transport. In contrast to combined marine-road transport, intermodal transportation involves block or public train services after the maritime logistics operations. In other words, intermodal is a transportation model wherein three distinct means of transport, i.e., maritime, railway and road transports are utilized. Intermodal transportation aims at transferring the loads from one location to another in a more effective, punctual and environment-friendly manner. In intermodal transportation, trailers or containers arrived by Ro-Ro vessels are loaded to block trains at the seaports in Europe and then transferred to rail stations. Finally, these loads are delivered to order countries by road freight transport. A featured intermodal transportation service provides lots of advantages such as timely delivery, cost-effectiveness and environmental awareness.

At seaports or railway stations, block trains and company-owned Ro-Ro vessels may arrive and depart in accordance with a fixed time schedule. In addition to these block train and Ro-Ro schedules, configuration of the intermodal transport network is also supposed to be fixed in this paper. The intermediate transshipments are also possible for both marine and rail transportation services based on the consolidation concept. In detail, some loads can be unloaded from the Ro-Ro vessels at a transshipment seaport and then send to an order country via road freight transport. Similarly, some extra loads can be loaded to the Ro-Ro vessels at a transshipment seaport and then transferred to any origin seaport in Turkey. In other words, intermediate stops may exist in marine transportation. Moreover, some loads can also be exchanged between the railway stations which are named as rail-rail exchange. In fact, some of the trailers can be unloaded from a block train at a transshipment railway station and loaded to another public train on the different rail line. The remaining trailers are delivered to order countries via road freight transport. On the other hand, these exchange operations may increase the transit time and total transport costs of the intermodal transportation system. For this reason, they should be performed rapidly and efficiently. The structure of the examined multi-mode load planning problem and configuration of the intermodal logistics network can be portrayed as in Figs. 3 and 4, respectively.

The proposed model also incorporates decisions on frequency of the Ro-Ro vessels and block train services. The objective is to find the least costly, less time consuming and the most environmental friendly load plans for the intermodal transportation system as shown by Fig. 5.

In the light of foregoing, the load plans can be constructed by solving the developed mixed-integer programming model in order to determine:

Table 2
An overview of the articles dealt with freight transportation planning problems in intermodal transportation networks.

Article	Problem definition	Problem characteristics	Freight planning	Fleet sizing	Empty repositions	Routing/ scheduling	Outsourcing	Environmental issues	Uncertainty conditions	Multiple objective	Solution methodology
Choong et al. (2002)	Empty container management	Static deterministic	–	–	✓	–	✓	–	–	–	Integer programming
Arnold et al. (2004)	Multimodal terminal location problem	Static deterministic	✓	–	–	–	–	–	–	–	Heuristic procedure based on shortest path
Chang (2008)	Intermodal freight routing problem	Static deterministic	✓	–	–	✓	–	–	–	✓	Lagrangian relaxation and decomposition
Puettmann and Stadtler (2010)	Decentralised intermodal planning	Dynamic stochastic	✓	–	✓	–	✓	–	✓	–	Collaborative planning approach
Bauer et al. (2010)	Intermodal scheduled service network design	Static deterministic	✓	✓	✓	✓	–	✓	–	✓	Integer linear programming
Verma and Verter (2010)	Planning rail–truck transportation	Static deterministic	✓	–	–	✓	–		–	✓	Iterative decomposition method
Yang et al. (2011)	Intermodal freight routing problem	Static deterministic	–	–	–	✓	–	–	–	✓	Goal programming model
Macharis et al. (2011)	Intermodal transport policy making	Dynamic stochastic	✓	–	–	✓	–	✓	✓	✓	Mathematical and simulation modeling
Bruns and Knust (2012)	Load planning problem of trains at terminals	Static deterministic	✓	✓	–	–	–	–	–	✓	Integer linear programming
Cho et al. (2012)	Multimodal routing with time windows	Static deterministic	–	-	-	✓	–	–	–	✓	Dynamic programming
Bierwirth et al. (2012)	Planning of mode & service selection	Static deterministic	✓	–	–	–	–	–	–	–	Mixed-integer programming
Verma et al. (2012)	Rail-truck intermodal transportation routing	Static deterministic	✓	–	✓	✓	–	–	–	✓	Tabu search based solution approach
Meisel et al. (2013)	Integrated production & transportation planning	Static deterministic	✓	–	–	–	–	✓	–	✓	Heuristic algorithm
Kalinina et al. (2013)	Matching of goods in intermodal transportation	Static Stochastic	✓	–	✓	✓	–	✓	✓	✓	Chance constrained programming
Garcia et al. (2013)	Assignment of containers & trucks to services	Dynamic Deterministic	✓	–	✓	✓	–	–	–	–	Integrated linear program and automated planning
Bhattacharya et al. (2014)	Strategic intermodal transport planning	Dynamic stochastic	✓	–	–	✓	–	–	✓	–	Mixed-integer programming
Assadipour et al. (2015)	Managing hazmat freight in intermodal transport	Static deterministic	✓	–	–	✓	–	–	–	✓	Genetic algorithm based solution method
Resat and Turkay (2015)	Intermodal transportation network design	Static deterministic	✓	–	–	✓	–	–	–	✓	MILP and augmented ε-constraint method
Li et al. (2015)	Container flow control in intermodal transportation	Dynamic stochastic	✓	–	–	✓	–	–	✓	✓	A receding horizon control approach

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Table 2 (*continued*)

Article	Problem definition	Problem characteristics	Freight planning	Fleet sizing	Empty repositions	Routing/ scheduling	Outsourcing	Environmental issues	Uncertainty conditions	Multiple objective	Solution methodology
Demir et al. (2015)	Intermodal service network design problem	Static Stochastic	✓	–	–	✓	–	✓	✓	–	Chance-constrained stochastic program
Ghane-Ezabadi and Vergara (2016)	Intermodal logistics network design with routing issues	Static deterministic	✓	–	–	✓	–	–	✓	–	Decomposition-based search algorithm
Febbraro et al. (2016)	Short/medium term intermodal transport planning	Dynamic stochastic	✓	–	–	✓	–	–	✓	–	Agent-based modeling
Present paper	Multi-mode load or intermodal freight transport planning	Static deterministic	✓	–	–	–	✓	✓	✓	✓	MILP and fuzzy multi-objective optimization

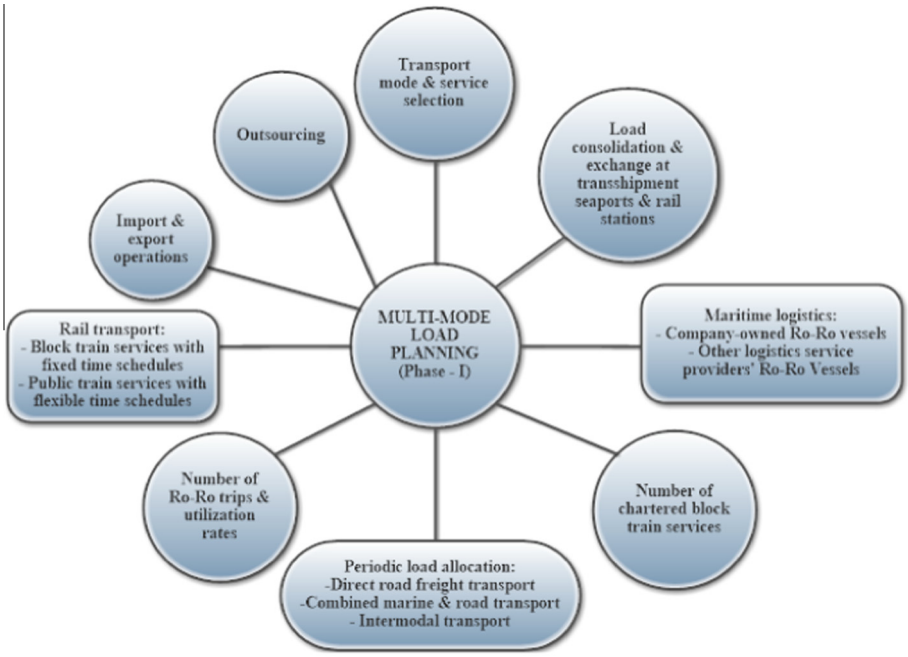


Fig. 3. Structure of the multi-mode load planning or intermodal freight transportation planning problem.

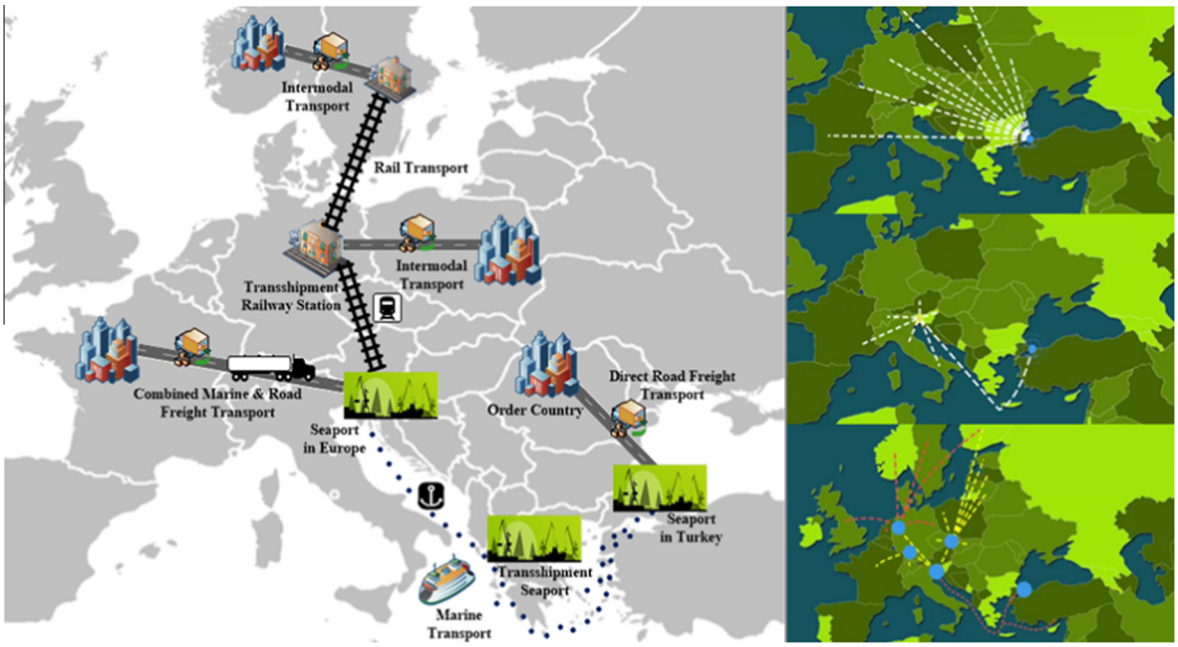


Fig. 4. Schematic depiction of the intermodal logistics network configuration.

- Effective mode selection and periodic load allocation to different transportation services,
- Composition of the direct road freight transport, combined marine-road transport, intermodal transportation and importation/exportation of loads by outsourcing option,
- Transportation amounts shipped by the company-owned Ro-Ro vessels and other logistics service providers' maritime services,
- The required number of marine cruises by the company-owned Ro-Ro vessels at each month,
- Utilization rates of the company-owned Ro-Ro vessels on different marine links,

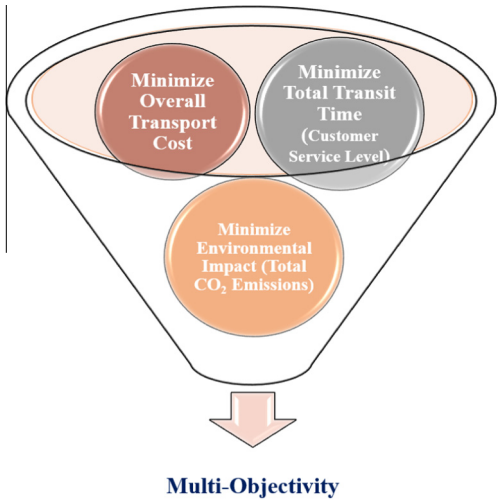


Fig. 5. The multi-objective nature of the proposed multi-mode load planning model.

- The required number of chartered block trains for both importation and exportation,
- The transportation amounts transferred by the public train services,
- Load transfers at the transshipment seaports and railway stations.

The proposed model is developed based on the following situations and assumptions. Indeed, these assumptions are made by the experts working at the intermodal freight transportation department of the company. These assumptions are also depend on some agreements on tariffs, working conditions, contracts with logistics service providers, customs clearance, railroad freight rates, etc.

- For both importation and exportation operations, domestic distribution services in Turkey are not taken into account by the proposed model. In other words, pre/end haulages in Turkey are not considered.
- The delayed or deferred customer transport demands are not allowed. All of the monthly freight demands of the order countries should be satisfied within the relevant time period.
- In maritime logistics, there is no capacity limitation on the quantity of transportation units (trailer or container) shipped by the other logistics service providers' Ro-Ro vessels.
- Outsourcing option has an unlimited transportation capacity. For this reason, the proposed model may have flexibility in freight transportation. Thus, it cannot produce infeasible solutions due to the hard freight demand satisfaction constraints when the demands exceed the transport capacities of the Ro-Ro vessels and trains. This option also covers the direct road freight transport and combined marine-road transport services.
- In rail transportation system, in contrast to the block trains with fixed time schedules, there is no limitation on the quantity of load units transferred by the public train services.
- Less than Truckload (LTL) is not permitted in the direct road freight transportation.
- Additional shipments can be loaded/unloaded at the transshipment seaports for only the company-owned Ro-Ro vessels during the maritime transportation operations.
- Monthly import and export freight demands of the order countries are exactly known and considered as deterministic.
- In Trieste seaport, importation and exportation operations can be carried out from/to Istanbul seaport by both of the company-owned and other logistics service providers' Ro-Ro vessels. On the other hand, trailers or containers can only be transferred by the other logistics service providers' Ro-Ro vessels between Trieste seaport and other origin seaports in Turkey, i.e., Izmir and Mersin seaports.
- For Sete seaport, transportation operations can be performed from/to Izmir origin seaport by only the company-owned Ro-Ro vessel. In fact, any import or export shipment cannot be performed by the other logistics service providers' Ro-Ro vessels.
- For import/export operations at a transshipment seaport, i.e., Lavrion, additional shipments can be loaded/unloaded during the maritime logistics service. Actually, this transshipment point is only connected with the marine link among Trieste and Istanbul seaports.
- The logistics company cannot organize any maritime service with its own Ro-Ro vessels between Istanbul and Toulon seaports. These shipments can be conducted by only the other logistics service providers' Ro-Ro vessels. Currently, it is not possible to organize any maritime service by company owned Ro-Ro vessels to Toulon seaport from Turkey.

- For the rail transportation, railway connections are just available from/to Trieste seaport in Europe. In detail, there are direct railway connections between Trieste seaport and some other railway terminals such as Ostrava, Munich, Ludwigshafen and Cologne. Actually, the rail transport operations between these terminals and Trieste seaport can only be achieved by chartering a whole block train with fixed time schedule.
- It is not possible to execute any rail transport operation between Trieste seaport and Munich rail station by the block trains. Alternatively, there are only public train services available between that seaport and rail station.
- Cologne station is a transshipment point on the railway network in Europe. In fact, some of the freights can be loaded or unloaded at this rail station and can be transferred by road freight transport so as to satisfy the freight demands of the surrounding order countries. The remaining loads are transferred to Malmö and Gothenburg rail stations by using public train services. Unfortunately, it is not possible to charter block trains between Malmö or Gothenburg rail stations and Cologne transshipment station.

According to the above explanations, a multi-objective, multi-mode, multi-period load planning model with simultaneous import/export transportation operations can be formulated through the following equations by using the network notation. Indices/sets, parameters and decision variables are introduced in Tables 3–5.

The first objective is to minimize the total transport cost of the overall intermodal transportation network which consists of marine transport cost per transportation unit by the company-owned or other logistics service providers' Ro-Ro vessels, rail transport costs per block train service, rail transport costs per transportation unit by public trains, outsourcing costs and road freight transportation cost of each transportation unit per kilometer (for loaded truck movements only).

(I) *Objective-1:* Minimize the overall transport cost throughout the intermodal logistics network:

$$\begin{aligned}
\textit{Minimize } Z_{Cost} = & \sum_{(i,j)\notin M_{ij}}\sum_{t\in T} R_{ijt}^{exp}\cdot Co_{ij}^{exp} + \sum_{(j,i)\notin M_{ji}}\sum_{t\in T} R_{jit}^{imp}\cdot Co_{ji}^{imp} + \sum_{(i,j)\notin C_{ij}}\sum_{t\in T} P_{ijt}^{exp}\cdot Cp_{ij}^{exp} + \sum_{(j,i)\notin C_{ji}}\sum_{t\in T} P_{jit}^{imp}\cdot Cp_{ji}^{imp} \\
& + \sum_{(j,k)\notin A_{jk}}\sum_{t\in T} S_{jkt}^{exp}\cdot Cb_{jk}^{exp} + \sum_{(k,j)\notin A_{kj}}\sum_{t\in T} S_{kjt}^{imp}\cdot Cb_{kj}^{imp} + \sum_{(j,k)\notin V_{jk}}\sum_{t\in T} Pt_{jkt}^{exp}\cdot Cp_{jk}^{exp} + \sum_{(k,j)\notin V_{kj}}\sum_{t\in T} Pt_{kjt}^{imp}\cdot Cp_{kj}^{imp} \\
& + \sum_{k^*\in G_k}\sum_{k'\in T_k}\sum_{t\in T} Pt_{k^*k't}^{imp}\cdot Cp_{k^*k'}^{imp} + \sum_{k'\in T_k}\sum_{k^*\in G_k}\sum_{t\in T} Pt_{k'k^*t}^{exp}\cdot Cp_{k'k^*}^{exp} + \sum_{i\in I}\sum_{j\in J}\sum_{l\in L}\sum_{t\in T} Y_{ijlt}^{exp}\cdot d_{jl}^{r}\cdot Cr \\
& + \sum_{l\in L}\sum_{j\in J}\sum_{i\in I}\sum_{t\in T} Y_{ljit}^{imp}\cdot d_{lj}^{r}\cdot Cr + \sum_{i\in I}\sum_{j\in J}\sum_{k\notin G_k}\sum_{l\in L}\sum_{t\in T} Z_{ijklt}^{exp}\cdot d_{kl}^{r}\cdot Cr + \sum_{l\in L}\sum_{k\notin G_k}\sum_{j\in J}\sum_{i\in I}\sum_{t\in T} Z_{lkjit}^{imp}\cdot d_{lk}^{r}\cdot Cr \\
& + \sum_{i\in I}\sum_{j\in J}\sum_{k^*\in G_k}\sum_{l\in G_l}\sum_{t\in T} Z_{ijk^*lt}^{exp}\cdot d_{k^*l}^{r}\cdot Cr + \sum_{l\in G_l}\sum_{k^*\in G_k}\sum_{j\in J}\sum_{i\in I}\sum_{t\in T} Z_{lk^*jit}^{imp}\cdot d_{lk^*}^{r}\cdot Cr + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} X_{ilt}^{exp}\cdot d_{il}^{r}\cdot Cr \\
& + \sum_{l\in L}\sum_{i\in I}\sum_{t\in T} X_{lit}^{imp}\cdot d_{li}^{r}\cdot Cr + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} O_{ilt}^{exp}\cdot SC_{ilt}^{exp} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} O_{lit}^{exp}\cdot SC_{lit}^{exp}
\end{aligned} \tag{1}$$

The second objective is to minimize the total transit time related to road, marine and railway transportation services. This objective can be associated with the customer satisfaction or service level. The transit time also includes loading/unloading times, travel times, times of consolidation operations at the transshipment points, customs clearance times and awaiting times at the seaports and rail stations. This objective contributes to timely delivery of loads to the destinations and satisfaction of customer demands as soon as possible. In addition to the road, marine and rail transit time calculations, the transfer times of the loads by the outsourcing option are also required. In this context, it is assumed that a specified large portion of the loads which are allocated to the outsourcing option is shipped by direct road freight transport. The remaining loads are transferred by combined marine-road freight transport.

(II) *Objective-2:* Minimize the total transit time or maximize customer service level:

$$\begin{aligned}
\textit{Minimize } Z_{Transit-time} = & \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} X_{ilt}^{exp}\cdot T_{il}^{r} + \sum_{l\in L}\sum_{i\in I}\sum_{t\in T} X_{lit}^{imp}\cdot T_{li}^{r} + \sum_{i\in I}\sum_{j\in J}\sum_{l\in L}\sum_{t\in T} Y_{ijlt}^{exp}\cdot (T_{ij}^{m}+T_{jl}^{r}) + \sum_{i\in I}\sum_{j\in J}\sum_{l\in L}\sum_{t\in T} Y_{ljit}^{imp}\cdot (T_{lj}^{r}+T_{ji}^{m}) \\
& + \sum_{i\in I}\sum_{j\in J}\sum_{k\notin G_k}\sum_{l\in L}\sum_{t\in T} Z_{ijklt}^{exp}\cdot (T_{ij}^{m}+T_{jk}^{t}+T_{kl}^{r}) + \sum_{l\in L}\sum_{k\notin G_k}\sum_{j\in J}\sum_{i\in I}\sum_{t\in T} Z_{lkjit}^{imp}\cdot (T_{lk}^{r}+T_{kj}^{t}+T_{ji}^{m}) \\
& + \sum_{i\in I}\sum_{j\in J}\sum_{k^*\in G_k}\sum_{l\in G_l}\sum_{t\in T} Z_{ijk^*lt}^{exp}\cdot (T_{ij}^{m}+T_{jk'}^{t}+T_{k'k^*}^{t}+T_{k^*l}^{r}) + \sum_{l\in G_l}\sum_{k^*\in G_k}\sum_{j\in J}\sum_{i\in I}\sum_{t\in T} Z_{lk^*jit}^{imp}\cdot (T_{lk^*}^{r}+T_{k^*k'}^{t}+T_{k'j}^{t}+T_{ji}^{m}) \\
& + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} O_{ilt}^{exp}\cdot [\varphi^{Out}\cdot T_{il}^{r} + (1-\varphi^{Out})\cdot \sum_{j\in J} T_{ij}^{m} + T_{jl}^{r}] + \sum_{l\in L}\sum_{i\in I}\sum_{t\in T} O_{lit}^{imp}\cdot [\varphi^{Out}\cdot T_{li}^{r} + (1-\varphi^{Out})\cdot \sum_{j\in J} T_{lj}^{r} + T_{ji}^{m}]
\end{aligned} \tag{2}$$

For the last objective, the logistics company aims at reducing the mileage on road with its intermodal logistics system and minimizing the carbon dioxide, hydrocarbon, particle and nitrogen emissions. The usage of intermodal transportation leads the majority of freights to marine and railway transports from the usual heavy lifter road freight transport. The CO_2 emission

Table 3
Logistics network notation.

Indices & sets:	
$i \in I$	seaports in Turkey for import/export marine transport operations, $I = \{1,\ldots,3\}$
$j, j', j^* \in J$	seaports in Europe for import/export marine transport operations, $J = \{1,\ldots,4\}$
$k, k', k^* \in K$	rail stations in Europe for import/export railway transport operations, $K = \{1,\ldots,6\}$
$l \in L$	order countries in Europe (aggregated customers), $L = \{1,\ldots,49\}$
$t \in T$	monthly time period, $T = \{1,\ldots,12\}$
Marine transport network subsets:	
$(i,j) \in B_{ij}$	set of marine links where import/export transport operations can be performed by both company-owned and other logistics service providers' Ro-Ro vessels,
$(i,j) \in C_{ij}$	set of marine links where import/export transport operations can be performed by only the company-owned Ro-Ro vessels,
$j \in D_j$	set of seaports in Europe which are directly connected with a railway station,
$j^* \in \gamma_j$	set of main origin-destination seaports in Europe connected with at least one transshipment seaport on the marine route,
$(i,j) \in M_{ij}$	set of marine links where import/export transport operations can be performed by only other logistics service providers' Ro-Ro vessels,
$j' \in T_j$	set of transshipment seaports in Europe where additional freights can be loaded/unloaded during the marine transport services,
Rail transport network subsets:	
$(j,k) \in A_{jk}$	set of rail lines where import/export operations can be performed by only public train services,
$k \in D_k$	set of railway stations which are directly connected with a seaport in Europe,
$k^* \in G_k$	set of main origin-destination rail stations connected with at least one transshipment rail station on the rail route,
$k' \in T_k$	set of transshipment stations which also have railway connections with other main rail stations,
$(j,k) \in V_{jk}$	set of rail lines where import/export operations can be performed by only chartering block trains,
$(j,k) \in W_{jk}$	set of rail lines where import/export operations can be performed by both block and public train services,
Road transport network subsets:	
$l \in G_l$	set of order countries that can be served by road freight transport from the main railway stations $k^* \in G_k$ which also involve railway transshipments

calculations are performed based on the transportation mode, marine/land route distances and load quantities. In order to compute the savings in CO_2 emissions resulting from the different transportation modes, emission figures of trucks compliant with Euro 5 standards are implemented (Ekol, 2016).

(III) *Objective-3:* Minimize the total CO_2 emissions by different transportation modes:

$$\begin{aligned}
\textit{Minimize } Z_{CO2} = {} & \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} X^{exp}_{ilt}\cdot d^{r}_{il}\cdot CO^{R}_{2} + \sum_{l\in L}\sum_{i\in I}\sum_{t\in T} X^{imp}_{lit}\cdot d^{r}_{li}\cdot CO^{R}_{2} + \sum_{i\in I}\sum_{j\in J}\sum_{l\in L}\sum_{t\in T} Y^{exp}_{ijlt}\cdot d^{r}_{jl}\cdot CO^{R}_{2} \\
& + \sum_{l\in L}\sum_{j\in J}\sum_{i\in I}\sum_{t\in T} Y^{imp}_{ljit}\cdot d^{r}_{lj}\cdot CO^{R}_{2} + \sum_{i\in I}\sum_{j\in J}\sum_{k\notin G_k}\sum_{l\in L}\sum_{t\in T} Z^{exp}_{ijklt}\cdot d^{r}_{kl}\cdot CO^{R}_{2} + \sum_{i\in I}\sum_{j\in J}\sum_{k^*\in G_k}\sum_{l\in G_l}\sum_{t\in T} Z^{exp}_{ijk^*lt}\cdot d^{r}_{k^*l}\cdot CO^{R}_{2} \\
& + \sum_{l\in L}\sum_{k\notin G_k}\sum_{j\in J}\sum_{i\in I}\sum_{t\in T} Z^{imp}_{lkjit}\cdot d^{r}_{lk}\cdot CO^{R}_{2} + \sum_{l\in G_l}\sum_{k^*\in G_k}\sum_{j\in J}\sum_{i\in I}\sum_{t\in T} Z^{imp}_{lk^*jit}\cdot d^{r}_{lk^*}\cdot CO^{R}_{2} + \sum_{(i,j)\notin M_{ij}}\sum_{t\in T} R^{exp}_{ijt}\cdot d^{m}_{ij}\cdot CO^{M}_{2} \\
& + \sum_{(j,i)\notin M_{ji}}\sum_{t\in T} R^{imp}_{jit}\cdot d^{m}_{ji}\cdot CO^{M}_{2} + \sum_{(i,j)\notin C_{ij}}\sum_{t\in T} P^{exp}_{ijt}\cdot d^{m}_{ij}\cdot CO^{M}_{2} + \sum_{(j,i)\notin C_{ji}}\sum_{t\in T} P^{imp}_{jit}\cdot d^{m}_{ji}\cdot CO^{M}_{2} \\
& + \sum_{(j,k)\notin A_{jk}}\sum_{t\in T} S^{exp}_{jkt}\cdot Cap^{exp}_{jk}\cdot d^{t}_{jk}\cdot CO^{T}_{2} + \sum_{(k,j)\notin A_{kj}}\sum_{t\in T} S^{imp}_{kjt}\cdot Cap^{imp}_{kj}\cdot d^{t}_{kj}\cdot CO^{T}_{2} + \sum_{(j,k)\notin V_{jk}}\sum_{t\in T} Pt^{exp}_{jkt}\cdot d^{t}_{jk}\cdot CO^{T}_{2} \\
& + \sum_{(k,j)\notin V_{kj}}\sum_{t\in T} Pt^{imp}_{kjt}\cdot d^{t}_{kj}\cdot CO^{T}_{2} + \sum_{k'\in T_k}\sum_{k^*\in G_k}\sum_{t\in T} Pt^{exp}_{k'k^*t}\cdot d^{t}_{k'k^*}\cdot CO^{T}_{2} + \sum_{k^*\in G_k}\sum_{k'\in T_k}\sum_{t\in T} Pt^{imp}_{k^*k't}\cdot d^{t}_{k^*k'}\cdot CO^{T}_{2} \\
& + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} O^{exp}_{ilt}\cdot\left[\varphi^{Out}\cdot d^{r}_{il}\cdot CO^{R}_{2} + (1-\varphi^{Out})\cdot\sum_{j\in J} d^{m}_{ij}\cdot CO^{M}_{2} + d^{r}_{jl}\cdot CO^{R}_{2}\right] \\
& + \sum_{l\in L}\sum_{i\in I}\sum_{t\in T} O^{imp}_{lit}\cdot\left[\varphi^{Out}\cdot d^{r}_{li}\cdot CO^{R}_{2} + (1-\varphi^{Out})\cdot\sum_{j\in J} d^{r}_{ij}\cdot CO^{R}_{2} + d^{m}_{ji}\cdot CO^{M}_{2}\right]
\end{aligned} \tag{3}$$

The constraints of the proposed mixed-integer programming model for an intermodal freight transportation or multi-mode load planning problem can be expressed by Eqs. (4)–(43):

i. Periodic load allocation to different modes of transport in order to satisfy freight demands of order countries by different transportation services:

$$X^{exp}_{ilt} + \sum_{j\in J} Y^{exp}_{ijlt} + \sum_{j\in D_j}\sum_{k\in K} Z^{exp}_{ijklt} + O^{exp}_{ilt} \geqslant De^{exp}_{ilt} \quad \forall i\in I, \forall l\in L, \forall t\in T \tag{4}$$

$$X^{imp}_{lit} + \sum_{j\in J} Y^{imp}_{ljit} + \sum_{k\in K}\sum_{j\in D_j} Z^{imp}_{lkjit} + O^{imp}_{lit} \geqslant De^{imp}_{lit} \quad \forall i\in I, \forall l\in L, \forall t\in T \tag{5}$$

Table 4
Model parameters.

Co_{ij}^{exp}	marine transport cost per TU of export operation by company owned Ro-Ro vessels on link $(i,j) \notin M_{ij}$
Co_{ji}^{imp}	marine transport cost per TU of import operation by company owned Ro-Ro vessels on link $(j,i) \notin M_{ji}$
Cp_{ij}^{exp}	marine cost per TU of exportation by other logistics service providers' Ro-Ro vessels on link $(i,j) \notin C_{ij}$
Cp_{ji}^{imp}	marine cost per TU of importation by other logistics service providers' Ro-Ro vessels on link $(j,i) \notin C_{ji}$
Cb_{jk}^{exp}	rail transport cost per block train chartered for export operation on line $(j,k) \notin A_{jk}$
Cb_{kj}^{imp}	rail transport cost per block train chartered for import operation on line $(k,j) \notin A_{kj}$
Cp_{jk}^{exp}	rail transport cost per TU of export operation by public trains on line $(j,k) \notin V_{jk}$
Cp_{kj}^{imp}	rail transport cost per TU of import operation by public trains on line $(j,k) \notin V_{jk}$
$Cp_{k'k^*}^{exp}$	rail transport cost per TU of export operation by public trains on line $(k',k^*)\|k' \in T_k$ *and* $k^* \in G_k$
$Cp_{k^*k'}^{imp}$	rail transport cost per TU of import operation by public trains on line $(k^*,k')\|k^* \in G_k$ *and* $k' \in T_k$
Cr	road freight transport cost per km of per TU by trucks for both import/export operations
Cap_{ij}^{exp}	transport capacity of a company-owned Ro-Ro vessel per trip of export operation on link $(i,j) \notin M_{ij}$
Cap_{ji}^{imp}	transport capacity of a company-owned Ro-Ro vessel per trip of import operation on link $(j,i) \notin M_{ji}$
$Cap_{jk}^{\exp}$	transport capacity of a block train per service of export operation on line $(j,k) \notin A_{jk}$ and $k \in D_k \cup T_k$
Cap_{kj}^{imp}	transport capacity of a block train per service of import operation on line $(k,j) \notin A_{kj}$ and $k \in D_k \cup T_k$
CO_2^M	amount of CO_2 emissions (in kg) by marine transport services per km of per TU
CO_2^R	amount of CO_2 emissions (in kg) by road freight transport per km of per TU
CO_2^T	amount of CO_2 emissions (in kg) by railway transport services per km of per TU
d_{il}^r	road travel distance between seaports in Turkey $i \in I$ and order country $l \in L$
d_{jl}^r	road travel distance between seaports in Europe $j \in J$ and order country $l \in L$
d_{kl}^r	road travel distance between rail stations $k \in D_k \cup T_k$ and order country $l \in L$
$d_{k^*l}^r$	road travel distance between rail stations $k^* \in G_k$ and order country $l \in G_l$
d_{ij}^m	marine travel distance between seaports in Turkey and Europe, $(i,j) \in M_{ij} \cup C_{ij} \cup B_{ij}$
d_{jk}^t	rail travel distance between seaports in Europe and rail stations, $(j,k) \in A_{jk} \cup V_{jk} \cup W_{jk}$ and $k \notin G_k$
$d_{k'k^*}^t$	rail travel distance between transshipment and main rail stations, $(k',k^*) \in A_{k'k^*}$, $k' \in T_k$ and $k^* \in G_k$
De_{ilt}^{exp}	freight demand of order country l for export operation from seaport in Turkey i in time period t
De_{lit}^{imp}	freight demand of order country l for import operation to seaport in Turkey i in time period t
LB_{jkt}^{exp}	lower bounds on number of block trains for exportation on line $(j,k) \notin A_{jk}$ and $k \in D_k \cup T_k$ in period t
LB_{kjt}^{imp}	lower bounds on number of block trains for importation on line $(k,j) \notin A_{kj}$ and $k \in D_k \cup T_k$ in period t
LM_{ijt}^{exp}	lower bounds on utilization rates of Ro-Ro vessels for exportation on marine link $(i,j) \notin M_{ij}$ in period t
LM_{jit}^{imp}	lower bounds on utilization rates of Ro-Ro vessels for importation on marine link $(j,i) \notin M_{ji}$ in period t
N_{ijt}^{exp}	maximum number of marine cruises that can be performed by company-owned Ro-Ro vessels for export operation on link $(i,j) \notin M_{ij}$ in time period t
N_{jit}^{imp}	maximum number of marine cruises that can be performed by company-owned Ro-Ro vessels for import operation on link $(j,i) \notin M_{ji}$ in time period t
SC_{ilt}^{exp}	outsourcing cost per TU of export operation from seaport in Turkey i to order country l in time period t
SC_{lit}^{imp}	outsourcing cost per TU of import operation from order country l to seaport in Turkey i in time period t
T_{ij}^m	marine transit time between seaports in Turkey and Europe, $(i,j) \in M_{ij} \cup C_{ij} \cup B_{ij}$
T_{jk}^t	rail transit time between seaports in Europe and rail stations, $(j,k) \in A_{jk} \cup V_{jk} \cup W_{jk}$ and $k \notin G_k$
$T_{jk'}^t$	rail transit time between seaports in Europe and transshipment rail stations, $k' \in T_k$ and $j \in J$
$T_{k'k^*}^t$	rail transit time between transshipment and main rail stations, $(k',k^*) \in A_{k'k^*}$, $k' \in T_k$ and $k^* \in G_k$
T_{il}^r	road transit time between seaports in Turkey $i \in I$ and order country $l \in L$
T_{jl}^r	road transit time between seaports in Europe $j \in J$ and order country $l \in L$
T_{kl}^r	road transit time between rail stations $k \in D_k \cup T_k$ and order country $l \in L$
$T_{k^*l}^r$	road transit time between rail stations $k^* \in G_k$ and order country $l \in G_l$
UB_{jkt}^{exp}	maximum number of block trains for export operation on line $(j,k) \notin A_{jk}$ and $k \in D_k \cup T_k$ in period t
UB_{kjt}^{imp}	maximum number of block trains for import operation on line $(k,j) \notin A_{kj}$ and $k \in D_k \cup T_k$ in period t
φ^{out}	specific portion of direct road freight transportation in outsourcing option

ii. The specific constraints on marine transportation services:

$$\sum_{l\in L} Y_{ijlt}^{exp} \leqslant N_{ij^*t}^{exp} \cdot Cap_{ij^*}^{exp} - \sum_{l\in L} Y_{ij^*lt}^{exp} - \sum_{l\in L}\sum_{k\in K} Z_{ij^*klt}^{exp} \quad \forall (i,j) \in C_{ij} \wedge j \in T_j, j^* \in \gamma_j, \forall t \in T \tag{6}$$

$$\sum_{l\in L} Y_{ljit}^{imp} \leqslant N_{j^*it}^{imp} \cdot Cap_{j^*i}^{imp} - \sum_{l\in L} Y_{lj^*it}^{imp} - \sum_{l\in L}\sum_{k\in K} Z_{lkj^*it}^{imp} \quad \forall (j,i) \in C_{ji} \wedge j \in T_j, j^* \in \gamma_j, \forall t \in T \tag{7}$$

Table 5
Decision variables of the proposed MILP model.

Different transport services and outsourcing:	
O_{ilt}^{exp}	export quantity of TUs transferred via outsourcing from seaport in Turkey i to order country l in period t
O_{lit}^{imp}	import quantity of TUs transferred via outsourcing from order country l to seaport in Turkey i in period t
X_{ilt}^{exp}	export quantity of TUs shipped by direct road freight transport on road network $(i,l) \in I \cup L$ in period t
X_{lit}^{imp}	import quantity of TUs shipped by direct road freight transport on road network $(l,i) \in L \cup I$ in period t
Y_{ijlt}^{exp}	export quantity of TUs shipped by integrated marine & road transport on marine link $(i,j) \in M_{ij} \cup C_{ij} \cup B_{ij}$ and road network $(j,l) \in J \cup L$ in time period t
Y_{ljit}^{imp}	import quantity of TUs shipped by integrated marine & road transport on marine link $(j,i) \in M_{ji} \cup C_{ji} \cup B_{ji}$ and road network $(l,j) \in L \cup J$ in time period t
Z_{ijklt}^{exp}	export quantity of TUs shipped by intermodal transport on marine link $(i,j) \in M_{ij} \cup C_{ij} \cup B_{ij}$ and rail line $(j,k) \in L_{jk} \cup C_{jk} \cup B_{jk}$ and then road network $(k,l) \in K \cup L$ in time period t
Z_{lkjit}^{imp}	import quantity of TUs shipped by intermodal transport on marine link $(j,i) \in M_{ji} \cup C_{ji} \cup B_{ji}$ and rail line $(k,j) \in L_{kj} \cup C_{kj} \cup B_{kj}$ and then road network $(l,k) \in L \cup K$ in time period t
Specific marine transport variables:	
MS_{ijt}^{exp}	number of cruises by company-owned Ro-Ro vessels for exportation on marine link $(i,j) \notin M_{ij}$ in period t
MS_{jit}^{imp}	number of cruises by company-owned Ro-Ro vessels for importation on marine link $(j,i) \notin M_{ji}$ in period t
P_{ijt}^{exp}	export quantity of TUs shipped by other logistics service providers' Ro-Ro vessels on marine link $(i,j) \notin C_{ij}$ in time period t
P_{jit}^{imp}	import quantity of TUs shipped by other logistics service providers' Ro-Ro vessels on marine link $(j,i) \notin C_{ji}$ in time period t
R_{ijt}^{exp}	export quantity of TUs shipped by company-owned Ro-Ro vessels on marine link $(i,j) \notin M_{ij}$ in period t
R_{jit}^{imp}	import quantity of TUs shipped by company-owned Ro-Ro vessels on marine link $(j,i) \notin M_{ji}$ in period t
U_{ijt}^{exp}	utilization rate of company-owned Ro-Ro vessels for exportation on marine link $(i,j) \notin M_{ij}$ in period t
U_{jit}^{imp}	utilization rate of company-owned Ro-Ro vessels for importation on marine link $(j,i) \notin M_{ji}$ in period t
Specific rail transport variables:	
Pt_{jkt}^{exp}	export quantity of TUs shipped by public train service on rail line $(j,k) \notin V_{jk}$ and $k \notin G_k$ in period t
Pt_{kjt}^{imp}	import quantity of TUs shipped by public train service on rail line $(k,j) \notin V_{kj}$ and $k \notin G_k$ in period t
$Pt_{k'k^*t}^{exp}$	export quantity of TUs shipped by public trains on rail line $(k',k^*)\|k' \in T_k$ and $k^* \in G_k$ in period t
$Pt_{k^*k't}^{imp}$	import quantity of TUs shipped by public trains on rail line $(k^*,k')\|k^* \in G_k$ and $k' \in T_k$ in period t
S_{jkt}^{exp}	number of chartered block trains for export operation on rail line $(j,k) \notin A_{jk}$ in time period t
S_{kjt}^{imp}	number of chartered block trains for import operation on rail line $(k,j) \notin A_{kj}$ in time period t

$$\sum_{l \in L} Y_{ijlt}^{exp} \leqslant N_{ijt}^{exp} \cdot Cap_{ij}^{exp} \quad \forall (i,j) \in C_{ij} \wedge j \notin D_j, \forall t \in T \tag{8}$$

$$\sum_{l \in L} Y_{ljit}^{imp} \leqslant N_{jit}^{imp} \cdot Cap_{ji}^{imp} \quad \forall (j,i) \in C_{ji} \wedge j \notin D_j, \forall t \in T \tag{9}$$

$$\sum_{l \in L} Y_{ijlt}^{exp} \leqslant N_{ijt}^{exp} \cdot Cap_{ij}^{exp} + P_{ijt}^{exp} \quad \forall (i,j) \in B_{ij} \wedge j \notin D_j, \forall t \in T \tag{10}$$

$$\sum_{l \in L} Y_{ljit}^{imp} \leqslant N_{jit}^{imp} \cdot Cap_{ji}^{imp} + P_{jit}^{imp} \quad \forall (j,i) \in B_{ji} \wedge j \notin D_j, \forall t \in T \tag{11}$$

$$\sum_{l \in L} Y_{ijlt}^{exp} = R_{ijt}^{exp} \quad \{\forall (i,j) \in C_{ij} \vee B_{ij}\} \wedge \{j \in T_j \vee j \notin D_j\}, \forall t \in T \tag{12}$$

$$\sum_{l \in L} Y_{ljit}^{imp} = R_{jit}^{imp} \quad \{\forall (j,i) \in C_{ji} \vee B_{ji}\} \wedge \{j \in T_j \vee j \notin D_j\}, \forall t \in T \tag{13}$$

$$\sum_{l \in L} Y_{ijlt}^{exp} + \sum_{k \in K} \sum_{l \in L} Z_{ijklt}^{exp} \leqslant N_{ijt}^{exp} \cdot Cap_{ij}^{exp} - \sum_{l \in L} Y_{ij'it}^{exp} + P_{ijt}^{exp} \quad \forall (i,j) \in C_{ij}, j \in D_j \wedge j \in \gamma_j, j' \in T_j, \forall t \in T \tag{14}$$

$$\sum_{l \in L} Y_{ljit}^{imp} + \sum_{k \in K} \sum_{l \in l} Z_{lkjit}^{imp} \leqslant N_{jit}^{imp} \cdot Cap_{ji}^{imp} - \sum_{l \in L} Y_{lj'it}^{imp} + p_{jit}^{imp} \quad (j,i) \in C_{ji}, j \in D_j \wedge j \in \gamma_j, j' \in \gamma_j, j' \in T_j, \forall t \in T \tag{15}$$

$$\sum_{l \in L} Y_{ijlt}^{exp} + \sum_{k \in K} \sum_{l \in L} Z_{ijklt}^{exp} = R_{ijt}^{exp} + P_{ijt}^{exp} \quad \forall (i,j) \in B_{ij} \vee C_{ij}, j \in D_j, \forall t \in T \tag{16}$$

$$\sum_{l \in L} Y_{ljit}^{imp} + \sum_{k \in K} \sum_{l \in l} Z_{lkjit}^{imp} = R_{jit}^{imp} + P_{jit}^{imp} \quad \forall (j,i) \in B_{ji} \vee C_{ji}, j \in D_j, \forall t \in T \tag{17}$$

$$\sum_{l\in L} Y^{exp}_{ijlt} + \sum_{k\in K}\sum_{l\in L} Z^{exp}_{ijklt} \leqslant P^{exp}_{ijt} \quad \forall (i,j) \notin C_{ij}, j \in D_j, \forall t \in T \tag{18}$$

$$\sum_{l\in L} Y^{imp}_{ljit} + \sum_{k\in K}\sum_{l\in L} Z^{imp}_{lkjit} \leqslant P^{imp}_{jit} \quad \forall (j,i) \notin C_{ji}, j \in D_j, \forall t \in T \tag{19}$$

$$\sum_{l\in L} Y^{exp}_{ijlt} \leqslant P^{exp}_{ijt} \quad \forall (i,j) \in M_{ij}, j \notin D_j, \forall t \in T \tag{20}$$

$$\sum_{l\in L} Y^{imp}_{ljit} \leqslant P^{imp}_{jit} \quad \forall (j,i) \in M_{ji}, j \notin D_j \forall t \in T \tag{21}$$

$$R^{exp}_{ijt} / Cap^{exp}_{ij} = MS^{exp}_{ijt} \quad \forall (i,j) \notin M_{ij}, j \notin \gamma_j, \forall t \in T \tag{22}$$

$$R^{imp}_{jit} / Cap^{imp}_{ji} = MS^{imp}_{jit} \quad \forall (j,i) \notin M_{ji}, j \notin \gamma_j, \forall t \in T \tag{23}$$

$$\left(R^{exp}_{ijt} + R^{exp}_{ij^*t}\right) \Big/ Cap^{exp}_{ij^*} = MS^{exp}_{ij^*t} \quad \forall (i,j) \notin M_{ij}, j \in T_j, j^* \in \gamma_j, \forall t \in T \tag{24}$$

$$\left(R^{imp}_{jit} + R^{imp}_{j^*it}\right) \Big/ Cap^{imp}_{j^*i} = MS^{imp}_{j^*it} \quad \forall (j,i) \notin M_{ji}, j \in T_j, j^* \in \gamma_j, \forall t \in T \tag{25}$$

$$U^{exp}_{ijt} = R^{exp}_{ijt} \Big/ \left(N^{exp}_{ijt} \cdot Cap^{exp}_{ij}\right) \quad \forall (i,j) \notin M_{ij}, \forall t \in T \tag{26}$$

$$U^{imp}_{jit} = R^{imp}_{jit} \Big/ \left(N^{imp}_{jit} \cdot Cap^{imp}_{ji}\right) \quad \forall (j,i) \notin M_{ji}, \forall t \in T \tag{27}$$

$$LM^{exp}_{ijt} \leqslant U^{exp}_{ijt} \text{ and } LM^{imp}_{jit} \leqslant U^{imp}_{jit} \quad \forall (i,j) \notin M_{ij}, \forall (j,i) \notin M_{ji}, \forall t \in T \tag{28}$$

iii. The specific constraints on railway transportation services:

$$\sum_{i\in I}\sum_{l\in L} Z^{exp}_{ijklt} \leqslant S^{exp}_{jkt} \cdot Cap^{exp}_{jk} \quad (j,k) \in V_{jk} \setminus \{T_k\}, \forall \in T \tag{29}$$

$$\sum_{i\in I}\sum_{l\in L} Z^{imp}_{lkjit} \leqslant S^{imp}_{kjt} \cdot Cap^{imp}_{kj} \quad (k,j) \in V_{kj} \setminus \{T_k\}, \forall \in T \tag{30}$$

$$\sum_{i\in I}\sum_{l\in L} Z^{exp}_{ijklt} \leqslant Pt^{exp}_{jkt} \quad (j,k) \in A_{jk}, \forall t \in T \tag{31}$$

$$\sum_{i\in I}\sum_{l\in L} Z^{imp}_{lkjit} \leqslant Pt^{imp}_{kjt} \quad (k,j) \in A_{kj}, \forall t \in T \tag{32}$$

$$\sum_{i\in I}\sum_{l\in L} Z^{exp}_{ijklt} \leqslant S^{exp}_{jkt} \cdot Cap^{exp}_{jk} + Pt^{exp}_{jkt} \quad (j,k) \in W_{jk} \setminus \{T_k\}, \forall t \in T \tag{33}$$

$$\sum_{i\in I}\sum_{l\in L} Z^{imp}_{lkjit} \leqslant S^{imp}_{kjt} \cdot Cap^{imp}_{kj} + Pt^{imp}_{kjt} \quad (k,j) \in W_{kj} \setminus \{T_k\}, \forall t \in T \tag{34}$$

$$\sum_{i\in I}\sum_{l\in L} Z^{exp}_{ijk'lt} + \sum_{i\in I}\sum_{l\in G_l}\sum_{k^*\in G_k} z^{exp}_{ijk^*lt} \leqslant S^{exp}_{jk't} \cdot Cap^{exp}_{jk'} \quad \forall j \in D_j, \quad \forall k' \in T_k \cap (j,k') \in V_{jk'}, \forall t \in T \tag{35}$$

$$\sum_{i\in I}\sum_{l\in L} Z^{imp}_{lk'jit} + \sum_{i\in I}\sum_{l\in G_l}\sum_{k^*\in G_k} Z^{imp}_{lk^*jit} \leqslant S^{imp}_{k'jt} \cdot Cap^{imp}_{k'j} \quad \forall j \in D_j, \forall k' \in T_k \cap (k',j) \in V_{k'j}, \forall t \in T \tag{36}$$

$$\sum_{i\in I}\sum_{j\in D_j}\sum_{l\in G_l} Z^{exp}_{ijk^*lt} \leqslant Pt^{exp}_{k'k^*t} \quad \forall k' \in T_k, \forall k^* \in G_k \cap (k',k^*) \in A_{k'k^*}, \forall t \in T \tag{37}$$

$$\sum_{i\in I}\sum_{j\in D_j}\sum_{l\in G_l} Z^{imp}_{lk^*jit} \leqslant Pt^{imp}_{k^*k't} \quad \forall k' \in T_k, \forall k \in G_k \cap (k^*,k') \in A_{k^*k'}, \forall t \in T \tag{38}$$

$$LB^{exp}_{jkt} \leqslant S^{exp}_{jkt} \leqslant UB^{exp}_{jkt} \text{ and } LB^{imp}_{kjt} \leqslant S^{imp}_{kjt} \leqslant UB^{imp}_{kjt} \quad \forall (k,j) \notin V_{kj}, \forall (j,k) \notin V_{jk}, \forall t \in T \tag{39}$$

$$\left| S^{exp}_{jkt} - S^{imp}_{kjt} \right| \leqslant 1 \quad \forall (k,j) \notin V_{kj}, \forall (j,k) \notin V_{jk}, \forall t \in T \tag{40}$$

$$\left| \sum_{t \in T} \left(S^{exp}_{jkt} - S^{imp}_{kjt} \right) \right| \leqslant 1 \quad \forall (k,j) \notin V_{kj}, \forall (j,k) \notin V_{jk} \tag{41}$$

iv. Non-negativity and integrality restrictions for decision variables:

$$0 \leqslant U^{imp}_{ijt} \leqslant 1 \text{ and } 0 \leqslant U^{exp}_{jit} \leqslant 1 \quad \forall (i,j) \notin M_{ij}, \forall (j,i) \notin M_{ji}, \forall t \in T \tag{42}$$

$$X^{imp}_{ilt},\ X^{exp}_{lit},\ Y^{imp}_{ijlt},\ Y^{exp}_{ljit},\ Z^{imp}_{ijklt},\ Z^{exp}_{lkjit}, O^{imp}_{ilt}, O^{imp}_{ilt}, Y^{imp}_{lj^*it}, Y^{exp}_{ij^*lt},\ Z^{imp}_{lkj^*it}, Z^{exp}_{ij^*klt}, R^{imp}_{jit}, R^{exp}_{ijt}, Y^{imp}_{lj'it}, Y^{exp}_{ij'lt}, S^{imp}_{k'jt}, S^{exp}_{jk't} P^{imp}_{ijt},$$
$$P^{exp}_{jit}, Pt^{imp}_{jkt}, Pt^{exp}_{kjt}, S^{imp}_{jkt}, S^{exp}_{kjt}, Pt^{imp}_{k^*k't}, Pt^{exp}_{k'k^*t} \geqslant 0 \text{ and integer} \tag{43}$$

Constraints (4) and (5) maintain that import and export freight demands of each order country have to be satisfied at each month through the different transportation services. According to constraints (6) and (7), transportation capacity of the company-owned Ro-Ro vessels should be sufficient for both importation and exportation operations between the origin and destination seaports. In other words, these constraints represent the capacity restrictions for load units that can be shipped by only the company owned Ro-Ro vessels on the specific marine links. It should also be noted here that the seaport which represents the origin or destination of the import/export transport operation has not any railway connection. In detail, the only alternative is the road freight transport before/after the marine transport service. Actually, these constraints are also valid for the transshipment seaports, i.e., Lavrion. Similarly, constraints (8) and (9) are also the capacity restrictions for the marine transport services which can only be performed by the company-owned Ro-Ro vessels from/to the seaports, i.e., Sete, which have no railway connection. Actually, these are the capacity limitations and balance equations for the marine links where the main seaports and transshipment seaports are included. Additionally, these marine capacity constraints may differentiate based on the railway connection of the seaport and type of the utilized Ro-Ro vessel. For instance, marine transport services can only be performed by the company-owned Ro-Ro vessels for the transshipment seaports. The constraints (10) and (11) are formulated for the marine transport operations which can be conducted by both company-owned and other logistics service providers' Ro-Ro vessels. Similar to the constraints (6)–(9), the seaports have not any railway connection. By the constraints (12) and (13), one can calculate the total amounts of load units transferred by only the company owned Ro-Ro vessels from/to the transshipment seaports or the other seaports which have no railway connection. Actually, these constraints are the balance equations for the seaports, i.e., Lavrion and Sete. Constraints (14) and (15) are also capacity restrictions for the seaports where import and export operations can be performed by both of the company-owned and other logistics service providers' Ro-Ro vessels. In contrast to the previously formulated constraints (6)–(13), it should be noted here that origin or destination seaport is directly connected with a railway station. Moreover, it is also supposed that the marine links related to the main seaport, i.e. Trieste, also incorporate at least one transshipment seaport such as Lavrion. In detail, this seaport is a transshipment point between the specified main seaports, i.e., Istanbul and Trieste. For this reason, additional shipments which are loaded/unloaded at this transshipment seaport should also be considered within the capacity of the company owned Ro-Ro vessel which is operated between the main origin and destination seaports. Similar to the constraints (12) and (13), constraints (16) and (17) enable the calculation of amounts of load units delivered by both company-owned and other logistics service providers' Ro-Ro vessels from/to the seaports which are directly connected with a railway station. In fact, these constraints correspond to the balance equations for the seaport such as Trieste which is connected with some railway stations, i.e., Ludwigshafen, Cologne, Munich, and Ostrava. According to the constraints (18) and (19), the quantities of import and export shipments in marine transportation can be computed. It should be mentioned here that these shipments are performed by only the other logistics service providers' Ro-Ro vessels. It is obvious that there should be a railway transportation service available before/after the associated seaports, i.e., Izmir-Trieste and Mersin-Trieste. Similarly, constraints (20) and (21) are also formulated for the marine transportation services which can only be performed by the other logistics service providers' Ro-Ro vessels by excluding the railway connections such as Istanbul – Toulon ports. Based on the constraints (22) and (23), the number of marine cruises that should be carried out by the company-owned Ro-Ro vessels on each marine link can be calculated. However, these constraints are only valid for the main seaports such as Izmir-Sete ports which do not contain any transshipment seaport on its route. Moreover, the decision variables related to these marine cruises are defined as real numbers since the loads of the other logistics companies can also be transported by the company-owned Ro-Ro vessels. Thus, specific percentage of the company's loads on each marine link at each time period can also be determined by making use of the continuous variables. In a similar way, the number of import/export marine cruises by the company-owned Ro-Ro vessels which also involve the transportation operations from/to transshipment seaports can be calculated by the constraints (24) and (25). Additionally, the utilization rates of the company-owned Ro-Ro vessels on each marine link at each month can be decided by means of the constraints (26) and (27). In detail, these capacity utilization rates are just valid for the company's loads which are only transported by its own Ro-Ro vessels. These rates can be computed by dividing the total number of monthly load units into the multiplication of maximum

number of available monthly cruises with the capacity of a Ro-Ro vessel. Since some agreements on marine tariffs and weekly fixed time schedules, there are lower and upper bounds on these utilization rates. These bounds can be expressed by the constraint set (28). Constraints (29) and (30) are the capacity restrictions on the load units that can be shipped by only the block trains on the railway network, i.e., Trieste-Ostrava, Trieste-Ludwigshafen. In constraints (31) and (32), the quantities of the import and export load units that can only be transferred by the public trains can be provided, i.e., Trieste-Munich. Similarly, capacity limitations are also formulated in constraints (33) and (34) for the railway stations where export/import transportation operations can be performed by both block and public train services. In all of the constraint sets from (29) to (34), transshipment railway stations, i.e., Cologne, are excluded while satisfying these capacity limitations. Because, transshipment stations are connected with some other railway stations such as Malmö and Gothenburg instead of a seaport, i.e., Trieste. In other words, constraints (35) and (36) represent the capacity limitations on the import/export transportation operations for the transshipment railway station. This station corresponds to a hub on the railway network and it is sited among the specified main railway stations. In fact, some of the load units which arrived to Cologne station by the block trains can be distributed to the surrounding order countries via road freight transport. Then, the remaining ones can be transferred to the other order countries via public trains. It should be emphasized here that the transportation operations between a transshipment railway station and its connected seaports have to be performed by using only the block trains. In other words, public train services are not available for these transportation operations between Cologne station and Trieste port. On the other hand, these operations between the transshipment railway station and some other relevant stations, i.e. Gothenburg and Malmö, can only be performed by using the public trains as formulated in constraints (37) and (38). Constraint (39) presents the monthly lower and upper bounds on the block trains services. Based on the prior experiences and expert opinions, the difference between the number of import and export block train services on each rail line at each month can be at most one. Actually, there should be a balance between the import and export railway transportation services in each month. These constraints are resulted from the monthly scheduling limitations and can be formulated as in Eq. (40). In other words, due to the fixed time schedules of the railway operations, there should be a balance between the monthly import and export frequency of the block train services. This balance is required for each month and also valid for the annual basis (see in Eq. (41)). According to constraint (41), the difference between the total annual import and export railway services of a block train on a specific rail line can become at most one. Finally, constraint (42) defines lower and upper bounds on the decision variables related to the capacity utilization rates of the company-owned Ro-Ro vessels. The constraint (43) preserves the non-negativity of the decision variables and maintains that some of these variables must take integer values.

4. A real-life application in a large-sized international logistics company

4.1. Data description and optimization results for single objective deterministic MILP model

In order to illustrate the usefulness, validity and practicality of the proposed model, a real-life application is presented in a large-sized international logistics company in Turkey. The company is the fastest growing Turkish Logistics Corporation in Europe with over 100,000 m^2 of own warehouses and 1200 multinational employees. In addition, it is listed among the pioneering suppliers of integrated logistics services in both Turkey and Europe. The company has several distribution centers with more than 450,000 m^2 indoor areas in Turkey and a fleet of over 4000 trucks, trailers and containers. The company also provides intermodal transportation services and intercontinental logistics solutions including maritime routes and railways to a substantial majority of the best known global firms and brands all around the world. In fact, thousands of containers full of materials, finished products, industrial and consumer goods are carried by its fleet at each month. Although each sector has different requirements, the company offers customized transportation solutions for various sectors such as automotive, fast moving consumer goods, food, medical, retail, technology, construction and textile industries. The turnovers of the company with respect to the different logistics services and sectors can be shown by Fig. 6. As mentioned previously, the company's intermodal transportation system is based on the shipments of trailers or containers from Istanbul, Izmir and Mersin seaports to the seaports of Trieste or Sete via company-owned Ro-Ro vessels.

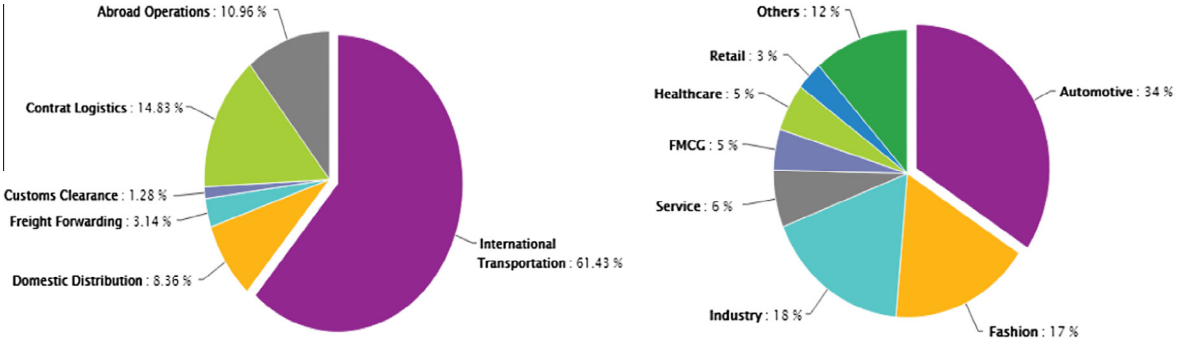


Fig. 6. The company turnovers with respect to different logistics services and sectors (EKOL, 2016).

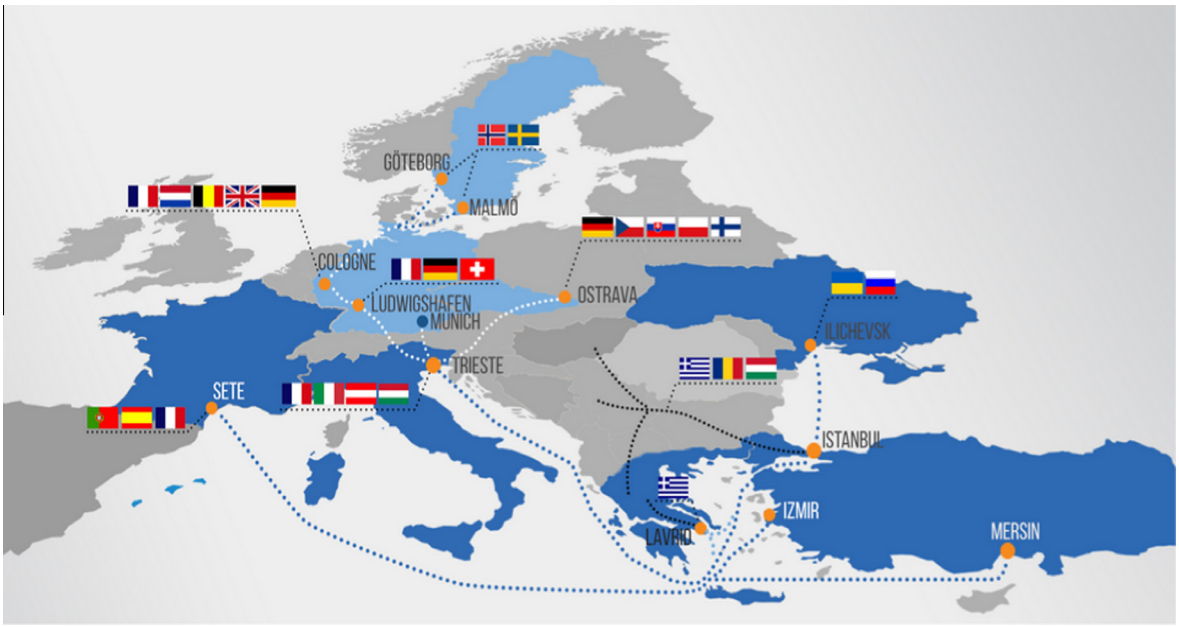


Fig. 7. Intermodal logistics network configuration of the real-life case study.

Arriving trailers and containers are unloaded from the block trains and transported to consolidation centers in Ostrava, Cologne and Ludwigshafen or to the final destinations via road freight transport. In the same manner, the loads are brought via road transport to Ludwigshafen or Cologne stations and consolidated there for transferring them to Trieste seaport via railways. Then, they are sent to Turkish origin seaports via the company owned Ro-Ro vessels in case of importation. The configuration of the real-life intermodal logistics network can be displayed by Fig. 7. In our case study, intermodal logistics network is composed of four seaports in Europe with a single transshipment seaport, three origin seaports in Turkey, three railway stations for block trains with a single transshipment railway station and finally three railway stations for only the public train services. There are also forty-nine order countries in Europe which correspond to the aggregated customers. The import/export freight demands of these order countries from/to Turkey should be satisfied. The planning horizon is set to one year and composed of twelve monthly periods. Before presenting the data used, structure of the examined real-life intermodal logistics network should be given. Tables 6 and 7 presented the marine and railway logistics network structures (or the subsets of marine and railway links) in terms of the proposed model formulation. Thus, the proposed model is reclaimed based on the following information that is gained from the intermodal freight transportation department of the company: 3 round trips per week take place between Istanbul origin seaport and Trieste port and a single round trip between Izmir and Sete seaports. Each trip's duration is 60 h and trips are realized with company owned Ro-Ro vessels which are equipped with the state-of-the-art technology and each vessel has a capacity of 249 mega-trailer or 45 ft container.

Each weekday, the company runs block trains on the railway network of Europe with approximately transit time of 6 days. Thus, it is possible to save many liters of diesel fuels and reduce carbon dioxide emissions compared to the consumptions in road freight transportation. The transportation capacity of a block train is defined as 32 trailers or containers per railway service. Additionally, there should be minimal 8 block train services available on each rail line at each month. The total number of annual freight demands of the order countries is approximately equal to 88,272 transportation units (trailers or containers). In detail, there are nearly 44,136 export and 42,656 import freight demands of the order countries in a year, respectively. The average monthly demands are also planned as 3554 and 3678 loads for the import and export transportation operations. The CO_2 emission values are computed on the kilometer basis as 0.918 kg/km for road freight transportation including a trailer with 23,800 kg load capacity. Furthermore, an average value of 0.357 kg/km is provided from Ecological Transport Information Tool (IFEU ECOTransIT) for marine CO_2 emissions. Finally, the CO_2 emission value

Table 6
Logistics network subsets related to the maritime transportation.

From/To	Istanbul			Izmir			Mersin			T_j	D_j	γ_j
	M_{ij}	C_{ij}	B_{ij}	M_{ij}	C_{ij}	B_{ij}	M_{ij}	C_{ij}	B_{ij}			
Trieste			⊗	⊗			⊗				⊗	⊗
Lavrion		⊗								⊗		
Sete					⊗							
Toulon	⊗											

Table 7
Railway line structure.

From/To	Trieste			T_k	D_k	G_k
	A_{kj}	V_{kj}	W_{kj}			
Ostrava		⊗			⊗	
Munich	⊗				⊗	
Ludwigshafen		⊗			⊗	
Cologne		⊗		⊗	⊗	
Malmö	⊗					⊗
Gothenburg	⊗					⊗

Table 8
Data ranges used in the case study.

Parameters	Range of values
Monthly import load demands of the order countries (TU/Month)	0–370
Monthly export load demands of the order countries (TU/Month)	0–366
Marine transport cost by the company owned Ro/Ro vessels (€/TU)	500–1475
Transportation capacity of the company owned Ro/Ro vessels (TU)	165–240
Marine transport cost by the other logistics service providers (€/TU)	647.73–918.3
The rail cost of per block train service (€/Train service)	20,121–23,328
Transportation capacity of the block trains (TU)	28–32
Rail transport cost by the public trains (€/TU)	685
Road freight transport cost per kilometer (€ TU/km)	1.20
Maximum cruise numbers by the company owned Ro/Ro vessels per month	4–14
Maximum train services by the block trains per month	12–41
The road distance between seaports in Turkey and order countries (km)	540–5070
The road transit time between seaports in Turkey and order countries (h)	27–254
The road distance between seaports in Europe and order countries (km)	50–4160
The road transit time between seaports in Europe and order countries (h)	2.5–208
The road distance between rail stations and order countries (km)	50–3015
The road transit time between rail stations and order countries (h)	2.5–150.8
The rail transit time between seaports and rail stations in Europe (h)	20–24
The marine transit time between seaports in Turkey and Europe (h)	17–96

Table 9
The payoff matrix and summary of the deterministic optimization results.

Objective functions	Total cost	Total transit time	Total CO_2 emission
Total transport cost (€)	**151890560.5**	8997322.46	106572584.5
Total transit time (h)	164507645.5	**8558320.15**	108593881.4
Total CO_2 emission (kg)	157078215.8	9004663.9	**100278315.6**
Model class	MILP		
Total number of variables	35729		
Integer variables	35592		
Total number of constraints	10728		
Total non-zero elements	174022		
Optimality gap (%)	0.0		
Total number of iterations	56107	2686	1370
CPU time (sec.)	94.73	19.18	19.04

is taken as 0.05 kg/km for the electrical trains which are usually used in Europe and are very slow in terms of average speed. The intervals of the other parameter values used in the real-life case study are given in Table 8.

When the model is run with these data through an optimization software LINGO 15.0 on an Intel Core i7 2 GHz IBM PC for each objective separately, the following deterministic optimization results and independent objective function values which are given in Table 9 are obtained. Actually, this table represents a payoff matrix and can be used as benchmarks to determine aspiration levels and max-min limits of each goal under uncertainty. In general, decision analysts can estimate the lower and upper limits of these goals based on their knowledge and experience. In addition, these limits can also be determined by using the payoff matrix which guaranteed feasibility of each fuzzy goal in the solution space. In detail, the main diagonal of this matrix denotes the individual best solutions for each objective. Then, lower and upper bounds can be designated from this matrix in order to be used in constructing the membership functions. Actually, these functions represent the satisfaction levels of the goals.

Moreover, optimal usage rates of different transportation modes and outsourcing option are given by Figs. 8–10 in case of single objective deterministic optimization. Actually, these rates are specified for both import and export transportation operations in an annual basis when the aforementioned objectives are optimized separately.

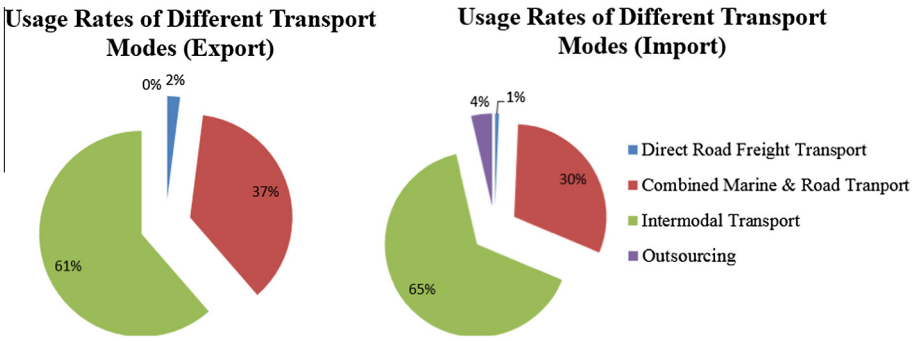


Fig. 8. Usage rates of different transportation modes and outsourcing: Minimize overall transport cost.

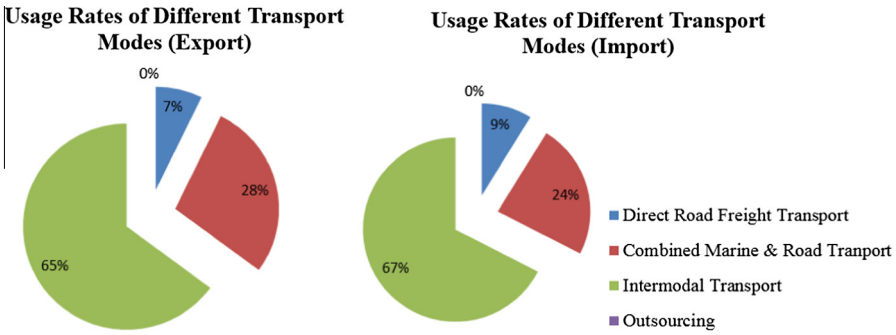


Fig. 9. Usage rates of different transportation modes and outsourcing: Minimize total transit time.

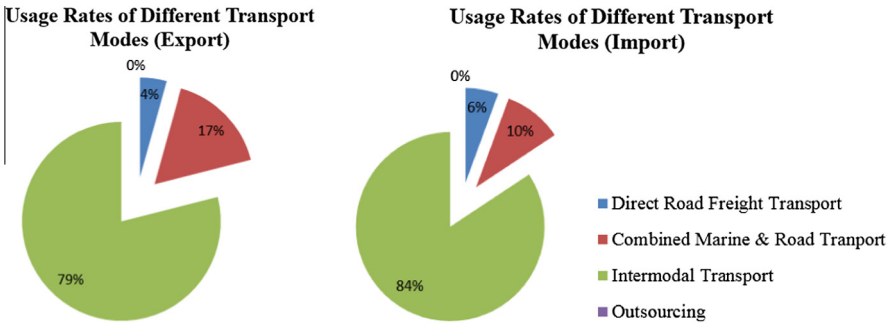


Fig. 10. Usage rates of different transportation modes and outsourcing: Minimize total CO_2 emission.

According to the deterministic optimization results, a large portion of the customers' freight demands should be satisfied by intermodal transportation for both importation and exportation. In fact, it is the most preferred transport option for the stated real-life load planning case. It can also be emphasized here that intermodal transportation is the most appropriate way of freighting in terms of economic, customer satisfaction and environmental issues. In other words, intermodal transportation sometimes contributes to all of these conflicting objectives. On the contrary to cost minimization where the marine transport cost constitutes the large portion of the overall transport cost, usage rate of the outsourcing option will increase when the transit time and environmental objectives are optimized. In such a situation, the usage rate of combined marine-road freight transport will decrease because of increases in the usage of intermodal transportation. Moreover, the usage of direct road freight transport will be relatively large due to its high-level of speed and flexibility in case of transit time minimization. According to Fig. 11, there is a balance among the monthly transport costs since there is no significant variation on the monthly import/export freight demands and total operational costs.

In terms of the cost objective, the variation of monthly usage rate of the outsourcing option for both import/export operations can also be demonstrated as in Fig. 12. There will be increases in import/export outsourcing operations at the beginning of the planning horizon. But, the amounts of loads transferred by the outsourcing will decrease towards the end of the year. Indeed, the imbalance between the import and export outsourcing quantities will disappear to the end of the planning

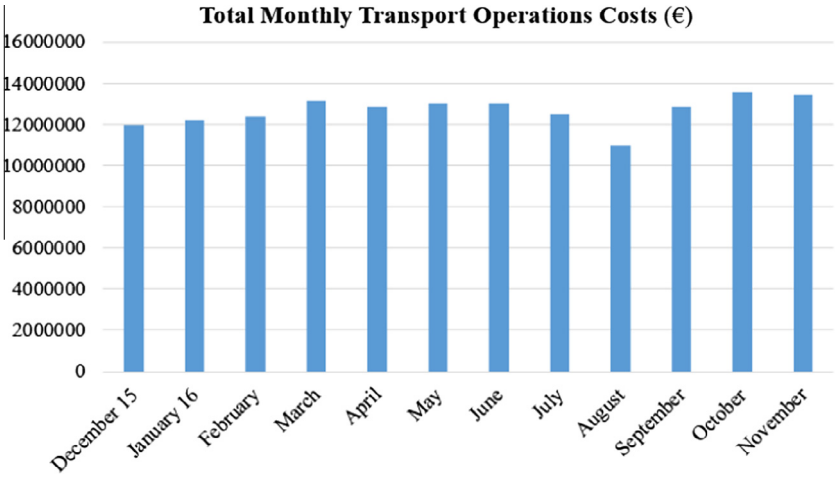


Fig. 11. The variation of monthly total transport costs.

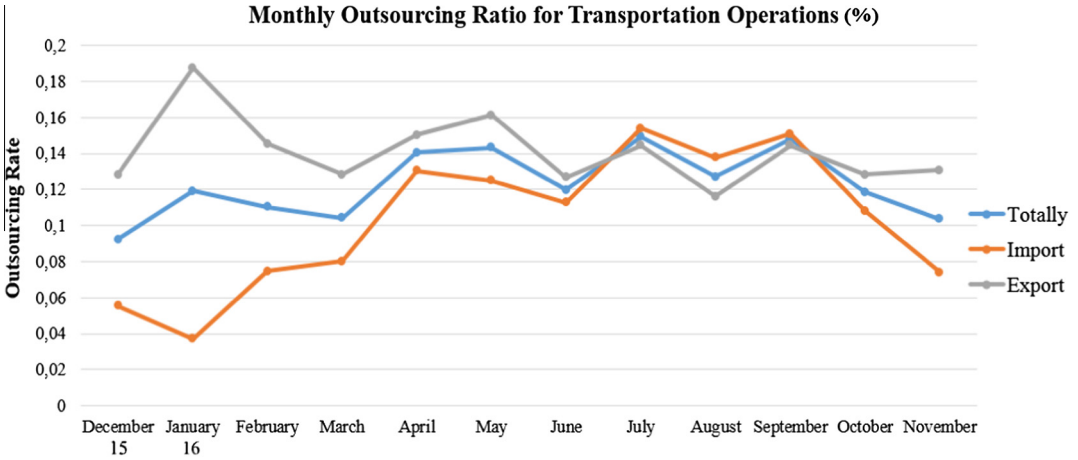


Fig. 12. The monthly variation of outsourcing ratios for import/export transportation operations.

period because of the increases in the usage of intermodal transportation. On the other hand, all of the import/export outsourcing quantities will shift to direct road freight transportation when the transit time objective is taken into account.

4.2. Multi-objective optimization via different approaches and comparison of the results

In order to generate compromise solutions to the decision analyst under crisp and uncertain environments, the following multiple objective optimization (MOO) approaches are chosen from the literature and applied to the proposed multi-mode load planning model:

- Compromise programming technique with different distance metrics (CP),
- Fuzzy goal programming approach with different importance and priorities (FGP-DIP),
- Weighted additive fuzzy goal programming approach (WFGP),
- Interactive fuzzy goal programming approach (IFGP).

In fact, all of these solution methodologies provided satisfactory results. However, since these methodologies have different characteristics and features, it is intended by this research that comparison of the results may be useful for the company experts under different circumstances. Additionally, different satisfactory results may offer meaningful managerial insights to the decision analysts. In Fig. 13, a decision support system framework is presented for the examined multi-mode load planning problem and its relevant solution algorithms. In the next subsections, the detailed explanations will be given for these utilized MOO approaches.

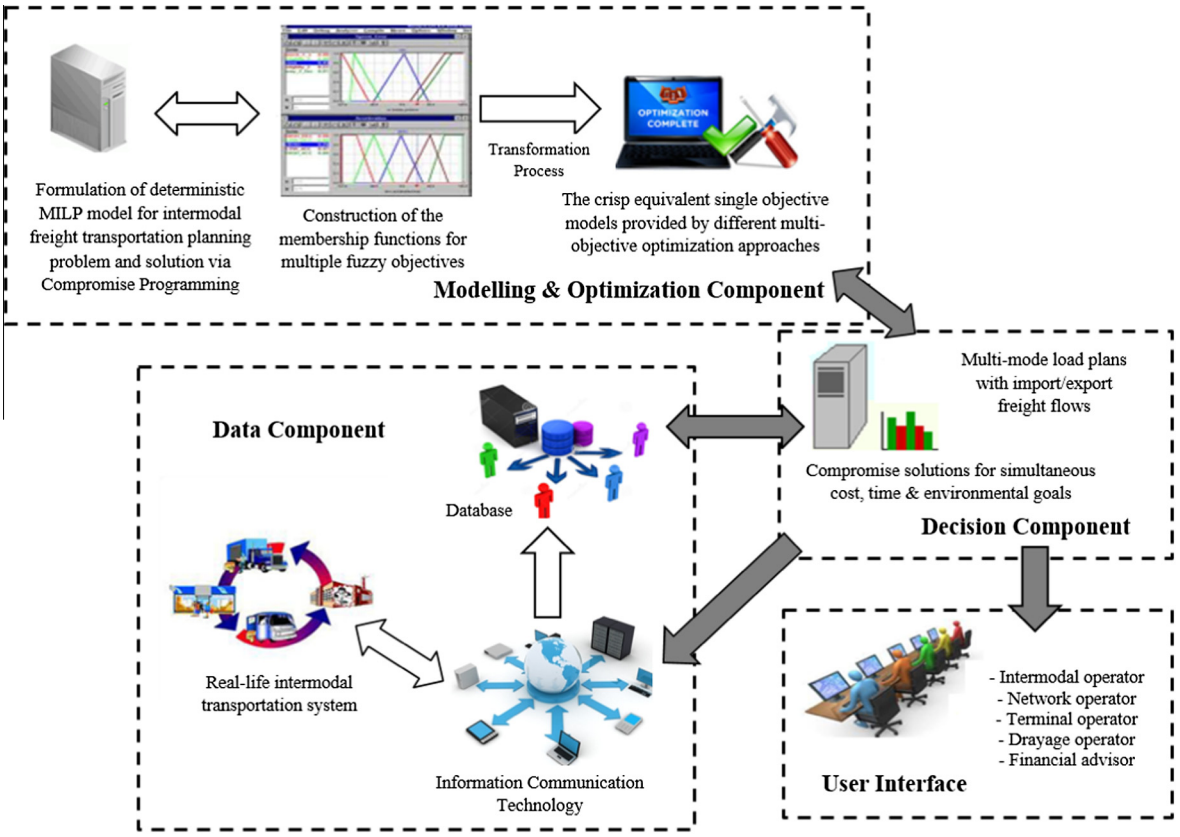


Fig. 13. A decision support system framework for multi-mode load planning problem and its solution algorithms.

4.2.1. Compromise programming approach

In order to solve the proposed multi-objective load planning model under certain environments, we employed compromise programming (CP) technique which was first introduced by Zeleny (1973). Apart from the latter fuzzy goal programming approaches, this technique does not require any subjective information from the decision analyst. In other words, the decision analyst does not need to release preferences and/or weights about the objectives. Additionally, this technique does not require any information on the target values of the goals. Moreover, relatively smaller number of computational effort is needed since it selects a unique optimal solution from a set of Pareto efficient solutions. For all of these reasons, it is an appropriate multi-objective optimization technique to find a compromise solution for the current load planning problem. This technique mainly depends on the choice of a unique solution which is the closest one to the ideal point. The closeness is measured by the following distance function (Gan et al., 1995):

$$d(x,y) = \left(\sum_{g=1}^{G} \left| f_g(x) - z_g^* \right|^p \right)^{1/p} \tag{44}$$

where $f_g(x); g = 1, \ldots, G$ are real valued objective functions and z_g^* represents the reference or ideal point. The value of parameter, p specifies a particular distance metric. For instance, when $p = 1$, it corresponds to Pythagorean distance between any given two points. When $p = 2$, it presents Euclidean distance. Actually, $p = 1$ and $p = \infty$ give the longest and shortest distances between these two points. The most widely used compromise solutions are for $p = 1$ and $p = \infty$ values because of their linear forms. However, other p values $1 < p < \infty$ need to solve non-linear programming models (Bilbao-Terol et al., 2006; Baykasoğlu et al., 2016). For example, the compromise program will be a quadratic program in the case of $p = 2$. Based on this information, the corresponding compromise program for the solution of multi-objective load planning model can be formulated as follows:

$$Minimize \left(\left| Z_{Cost} - Z_1^* \right|^p + \left| Z_{Transit-time} - Z_2^* \right|^p + \left| Z_{CO2} - Z_3^* \right|^p \right)^{1/p} \tag{45}$$

In order to reach the compromise solutions, the following linear and non-linear (quadratic) compromise programs are solved via LINGO 15.0 optimization software. Before solving these compromise programs, reference or ideal points should be found since this technique aims at minimizing the total distances. To this end, the proposed model is solved for each objective

function separately. Then, the following ideal points are achieved as $Z_1^* = 1518900560.5$, $Z_2^* = 8558320.15$ and $Z_3^* = 100278315.6$. These ideal points are also presented in payoff Table 9.

$$Min\ Z_C = \left|Z_{Cost} - Z_1^*\right| + \left|Z_{Transit-time} - Z_2^*\right| + \left|Z_{CO2} - Z_3^*\right| \quad p = 1, Pythagorean\ distance \tag{46}$$

$$Min\ Z_C = \sqrt{\left(Z_{Cost} - Z_1^*\right)^2 + \left(Z_{Transit-time} - Z_2^*\right)^2 + \left(Z_{CO2} - Z_3^*\right)^2} \quad p = 2, Euclidean\ distance \tag{47}$$

$$Min\ Z_C = Max\left(\left|Z_{Cost} - Z_1^*\right|, \left|Z_{Transit-time} - Z_2^*\right|, \left|Z_{CO2} - Z_3^*\right|\right) \quad p = \infty, Chebyshev\ distance \tag{48}$$

Since objective function of the compromise program with Pythagorean distance metric given in Eq. (46) involves summation of the absolute value terms, this non-linear objective function should be linearized as follows (Baker, 2015):

$$\begin{cases} Minimize\ Z_1^+ + Z_1^- + Z_2^+ + Z_2^- + Z_3^+ + Z_3^- \\ Z_{Cost} - Z_1^* + Z_1^+ - Z_1^- = 0 \\ Z_{Transit-time} - Z_2^* + Z_2^+ - Z_2^- = 0 \\ Z_{CO2} - Z_3^* + Z_3^+ - Z_3^- = 0 \\ Constraint\ Set(4) - (43) \\ Z_1^+, Z_1^-, Z_2^+, Z_2^-, Z_3^+\ and\ Z_3^- \geqslant 0 \end{cases} \tag{49}$$

where Z_1^+, Z_1^-, Z_2^+, Z_2^-, Z_3^+ and Z_3^- present the auxiliary variables to account for the positive and negative deviations. Similarly, due to the usage of max operator for the absolute values of these deviations, objective function of the compromise program with Chebyshev distance metric ($p = \infty$) can also be linearized as follows (Mangasarian, 2007, 2013):

$$\begin{cases} Minimize\ \delta \\ Z_{Cost} - Z_1^* + Z_1^+ - Z_1^- = 0 \\ Z_{Transit-time} - Z_2^* + Z_2^+ - Z_2^- = 0 \\ Z_{CO2} - Z_3^* + Z_3^+ - Z_3^- = 0 \\ Z_1^+ + Z_1^- \leq \delta \\ Z_1^+ + Z_1^- \geq \delta - (1 - \theta_1) \cdot M \\ Z_2^+ + Z_2^- \leq \delta \\ Z_2^+ + Z_2^- \geq \delta - (1 - \theta_2) \cdot M \\ Z_3^+ + Z_3^- \leq \delta \\ Z_3^+ + Z_3^- \geq \delta - (1 - \theta_3) \cdot M \\ \theta_1 + \theta_2 + \theta_3 \geq 1\ \theta_1,\ \theta_2\ and\ \theta_3 \in \{0, 1\} \\ Constraint\ Set\ (4) - (43)\ and\ Z_1^+, Z_1^-, Z_2^+, Z_2^-, Z_3^+, Z_3^- \geq 0 \end{cases} \tag{50}$$

where δ represents an auxiliary variable which is set to the maximum absolute value of the deviations. θ_1, θ_2 and $\theta_3 \in \{0, 1\}$ are binary variables to transform the maximum operator into linear form. Also, M is a huge positive number. Finally, solution of the multi-objective load planning model with deterministic targets is provided as in Table 14 by making use of the compromise programming technique with different distance metrics. Moreover, CP approach presents the following usage rates of different transportation modes as given by Figs. 14–16.

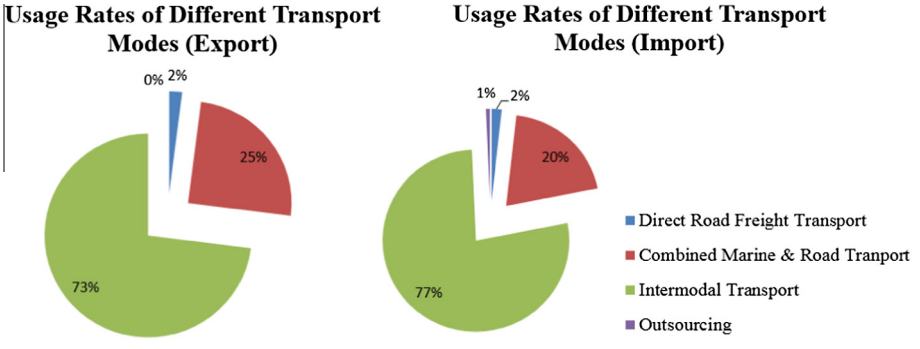


Fig. 14. Usage rates of transportation modes and outsourcing via CP with Pythagorean distance metric ($p = 1$).

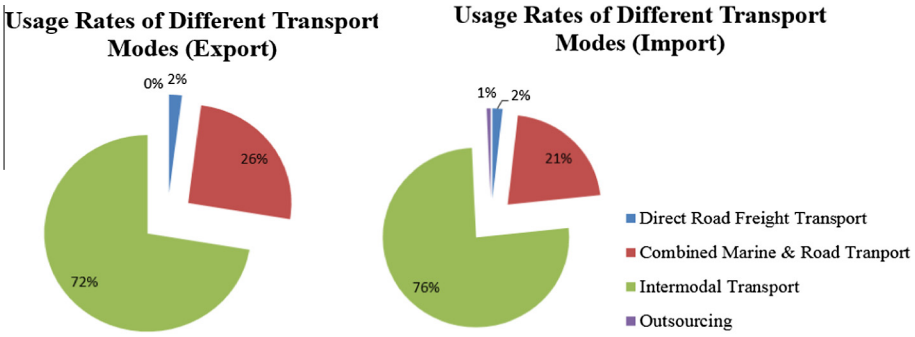


Fig. 15. Usage rates of transportation modes and outsourcing via CP with Euclidean distance metric ($p = 2$).

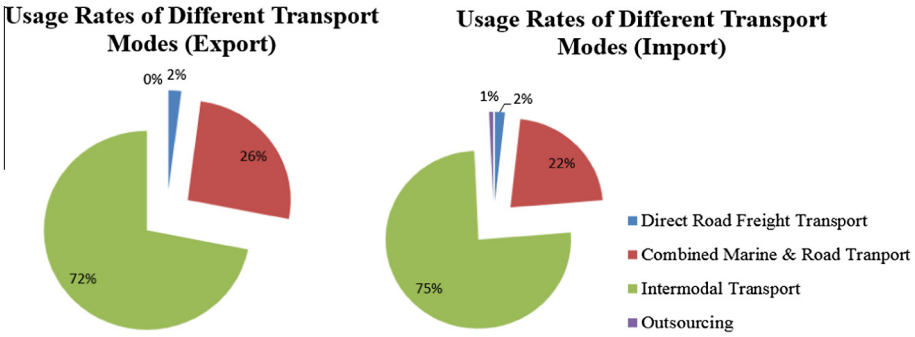


Fig. 16. Usage rates of transportation modes and outsourcing via CP with Chebyshev distance metric ($p = \infty$).

The multi-objective computational analysis has shown that similar results can be obtained from the application of CP approaches with different distance metrics. It is also obvious that the usage rate of intermodal transportation will increase whereas the rates of direct road freight transport and outsourcing will decrease when compared to the single objective deterministic optimization results. It should also be noted here that CP approach provided comparatively better results in terms total transport cost over the latter fuzzy goal programming based approaches. Furthermore, the best environmental results can also be yielded by CP approach with Pythagorean distance metric as clearly seen from Fig. 27. However, satisfactory solutions cannot be yielded from the CP approaches in terms of customer satisfaction. Because, the generated compromise solutions have relatively larger total transit times than the fuzzy goal programming based approaches. Lastly, it should be emphasized here that uncertainty related to the aspiration levels of the goals is not considered by CP approach. Because, fuzzy distance measures have complex structure in fuzzy version of the CP approach.

4.2.2. Fuzzy goal programming approach with different importance & priorities

The assessment and measurement of the total transport cost, transit time and CO_2 emission objectives explicitly are challenging tasks for the decision analyst and require a great deal of human perception. Because, some information and data may be incomplete or unobtainable for the coefficients into these objective functions. Therefore, objective values or aspiration levels of these goals may also become fuzzy in character. Therefore, importance weights, satisfaction sequence or achievement order of these objectives should be evaluated linguistically. In order to deal with this uncertainty embedded into these aspiration levels, Chen and Tsai (2001) proposed a fuzzy goal programming approach by considering different importance and priorities. In detail, they incorporated the different importance and preemptive priority structure into a single formulation by using an additive model. On the other hand, determination of a desirable achievement degree related to each goal is a difficult task for the decision analyst under such a fuzzy environment. Therefore, the most preferred method is to use linguistic terms/-expressions for identifying these desirable achievement degrees imprecisely. The linguistic term can be defined as a variable whose values are words or phrase in natural or artificial language (Jamalnia and Soukhakian, 2009). In this regard, a fuzzy linguistic variable is generally characterized via fuzzy numbers (Chang, 1996). The imprecise or vague nature of these linguistic assessments about the importance of different goals can be characterized by the following triangular fuzzy numbers given in Table 10. Furthermore, membership functions of these linguistic variables for the importance of different fuzzy goals are also depicted in Fig. 17. In this figure, $\mu_L(\beta)$ represents the membership function of each linguistic variable, where $\mu_L(\beta) \in [0, 1]$ and β display a variable which takes an achievement degree in the interval of $[\beta_{min}, \beta_{max}]$, $0 \leqslant \beta_{min} \leqslant \beta_{max} \leqslant 1$.

In order to calculate value of the desirable achievement degree for each fuzzy goal, Liou and Wang's (1992) defuzzification based ranking method is employed. This method ranks the fuzzy numbers in a group decision making environment where the index of optimism of each group member is different. In this approach, the total integral value $I_T^{\beta}(\widetilde{A})$ of a triangular

Table 10
Linguistic variables for the importance weights of different goals under uncertainty.

Linguistic variables	Triangular fuzzy numbers
Very low important (VLI)	(0, 0, 10%)
Low important (LI)	(5%, 15%, 25%)
Somewhat low important (SLI)	(20%, 32.5%, 45%)
Medium (M)	(40%, 50%, 60%)
Somewhat high important (SHI)	(55%, 67.5%, 80%)
High important (HI)	(75%, 85%, 95%)
Very high important (VHI)	(90%, 100%, 100%)

fuzzy number $\tilde{A} = (a, b, c)$ can be calculated as follows based on the index of optimism, $\beta \in [0, 1]$. This index reflects the decision maker's optimistic attitude.

$$I_T^{\beta}(\tilde{A}) = \frac{1}{2}[\beta \cdot c + b + (1 - \beta)a] \tag{51}$$

The total integral value represents a pessimistic decision maker when β is equal to "0". Additionally, it is suitable for moderate and optimistic decision makers where the value of β is set to "0.5" and "1.0", respectively. In the light of this information, value of this total integral value, $I_T^{\beta}(\tilde{A})$ can be used as a desirable achievement degree (θ_{ik}) of kth fuzzy goal for ith decision maker. In this research, the linguistic evaluations about the importance of the fuzzy goals are performed as shown in Table 11 by five different experts of the logistics company who have different index of optimisms. Actually, all of these experts are responsible for quite different intermodal transportation planning activities.

After obtaining the linguistic assessments from the company experts, the individual desirable achievement degree of each goal for each expert can be calculated as given in Table 12 by using the fuzzy ranking approach of Liou and Wang (1992). In fact, the total integral values of these linguistic variables are computed for each objective function by using Eq. (51). Briefly, the triangular fuzzy numbers associated with these linguistic evaluations are defuzzified according to the optimism index of company experts. Before calculating the overall priority (or aggregated desirable achievement degree) of each fuzzy goal, hierarchy between the company experts and importance weights of the group members should be determined.

Actually, there may not an exact hierarchy among the decision makers in most of the real-life settings. Furthermore, the relative importance or hierarchy of the group members may change over time. Thus, this hierarchy may also have some sort of ambiguity. For this purpose, the company experts are ranked in a fuzzy way in this study. Then, the relative importance or fuzzy weights of the group members are determined by using a subjective method namely "*Rank Reciprocal*". This method is solely based on the preference information given by an expert (Malczewski, 1999). After calculation of the reciprocal fuzzy weights of the company experts, they are normalized by using the following normalization procedure for fuzzy numbers (Fenton and Wang, 2006):

$$\tilde{W}_i = \left\{ \left(\frac{w_{i1}}{M}, \frac{w_{i2}}{M}, \frac{w_{i3}}{M} \right), \quad M = \max_i \{w_{i3}\} \right. \tag{52}$$

In order to calculate the fuzzy weights of the group members, the hierarchy between the company experts is determined by the board chairman of the logistics company. In fact, these experts are ranked in fuzzy way as $\tilde{1}, \tilde{2}, \tilde{3}, \tilde{4}$ and $\tilde{5}$ and then fuzzy hierarchy among the group members is depicted by the following Fig. 18.

After assigning the fuzzy ranks and calculating the reciprocal fuzzy weights, the normalized importance weights of the company experts are provided as in Table 13. Subsequently, the normalized fuzzy importance weights of the group members

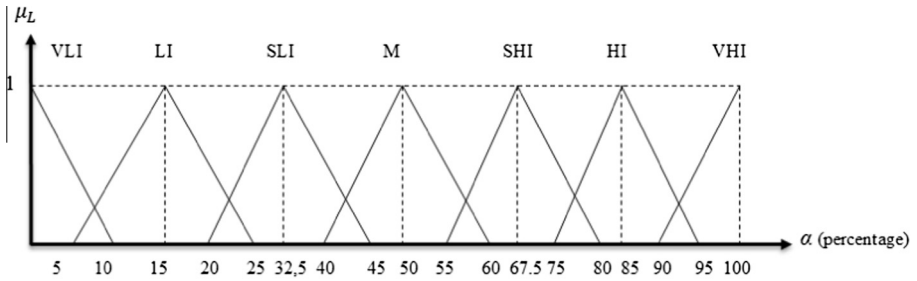


Fig. 17. Membership functions of the linguistic variables (Jamalnia and Soukhakian, 2009).

Table 11
Linguistic evaluations for relative importance of the fuzzy goals by different company experts.

Decision makers	Drayage operator	Terminal operator	Financial advisor	Network operator	Intermodal operator
Index of optimism	Pessimistic	Optimistic	Moderate	Optimistic	Moderate
Total transport cost	VHI	M	VHI	SHI	SHI
Total transit time	VHI	HI	M	HI	SHI
Total CO_2 emission	M	SLI	SLI	SHI	VHI

Table 12
Individual desirable achievement degree of each fuzzy goal for each company expert.

Decision makers	Drayage operator	Terminal operator	Financial advisor	Network operator	Intermodal operator
Index of optimism	Pessimistic	Optimistic	Moderate	Optimistic	Moderate
Total transport cost	0.95	0.55	0.975	0.7375	0.675
Total transit time	0.95	0.9	0.5	0.9	0.675
Total CO_2 emission	0.45	0.3875	0.325	0.7375	0.975

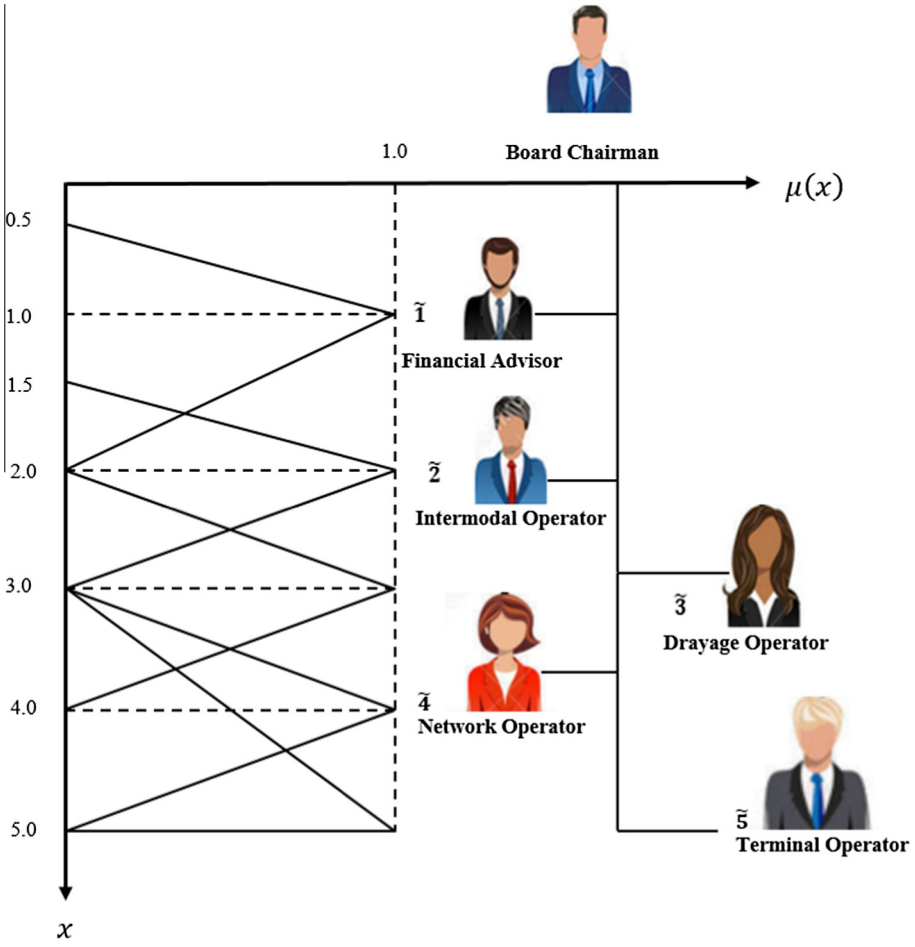


Fig. 18. The fuzzy hierarchy between the group members in an international logistics company.

are defuzzified via the fuzzy ranking approach of Liou and Wang (1992) based on the optimism index of the board chairman. In this research, it is supposed that the board chairman of the logistics company has a moderate optimistic attitude.

After determining the importance weights of the company experts, the overall priority of each fuzzy goal can be calculated by the following Eq. (53) based on the weighted geometric mean of individual desirable achievement degrees (Subulan et al., 2015a):

Table 13
Determination of importance weights of different company experts via rank reciprocal method.

Decision makers	Drayage operator	Terminal operator	Financial advisor	Network operator	Intermodal operator
Fuzzy rank	(2, 3, 4)	(3, 5, 5)	(0.5, 1, 2)	(3, 4, 5)	(1.5, 2, 3)
Reciprocal fuzzy weights	(0.25, 0.33, 0.5)	(0.2, 0.2, 0.33)	(0.5, 1, 2)	(0.2, 0.25, 0.33)	(0.33, 0.5, 0.67)
Normalized fuzzy weights	(0.125, 0.16, 0.25)	(0.1, 0.1, 0.16)	(0.25, 0.5, 1)	(0.1, 0.125, 0.16)	(0.16, 0.25, 0.33)
Defuzzified weights	0.177	0.116	0.563	0.129	0.25

$$\overline{\theta_k} = \left(\prod_{i=1}^{n} \theta_{ik}^{w_i}\right)^{1 \Big/ \sum_{i=1}^{n} w_i} \tag{53}$$

where θ_k is overall priority of the kth fuzzy goal. θ_{ik} is the individual desirable achievement degree of ith decision maker for the kth fuzzy goal. w_i is the importance weight of the ith group member. In summary, final desirable achievement degrees are computed as $\overline{\theta_1} = 0.83, \overline{\theta_2} = 0.653$ and $\overline{\theta_3} = 0.471$ for the total transport cost, transit time and CO_2 emission objectives, respectively. After that, the previously given payoff matrix in Table 9 is also used to specify the lower and upper limits of each fuzzy goal. To quantify the fuzzy aspiration levels of these goals, linear and continuous membership functions are found to be appropriate. Lastly, the membership functions of these fuzzy goals and their analytical definitions can be constructed as displayed in Fig. 19 by making use of these limits (Zimmermann, 1976). Finally, crisp equivalent form of the fuzzy multi-objective load planning model is formulated in Eq. (54) via the method of Chen and Tsai (2001). In this formulation, Z_k^U represents the upper bounds of the kth objective function which are derived from the payoff matrix. The objective of this crisp equivalent model is to maximize the summation of desirable achievement degrees for all fuzzy goals. These desirable achievement degrees or overall priorities reflect the relative importance of the fuzzy goals. Thus, more important goals will have larger achievement degrees. The model constraints ensure that achievement degrees of the fuzzy goals must be at least pre-specified overall priority levels. In the solution phase, infeasibility problem may occur when the overall priority levels are set too high. In this situation, membership functions of the fuzzy goals and the model parameters can be modified with respect to the company experts' preferences. Alternatively, the achievement degrees can be reduced rationally (Chen and Tsai, 2001; Subulan et al., 2015a). As a result, optimal usage rates of different transportation modes and outsourcing can be obtained as in Fig. 20 when the fuzzy goal programming approach of Chen and Tsai (2001) is applied. Although this approach offers satisfactory solutions to the cost and transit time objectives, it causes significantly high CO_2 emission values when compared to the other MOO approaches. The computational results of this fuzzy goal programming approach highlighted that decreases in the usage rate of intermodal transportation and increases in the outsourcing ratio will cause extra costs and high environmental impact. On the other hand, this provides transit time reduction which increases the customer satisfaction level.

$$\begin{cases} \textit{Maximize} \sum_{k=1}^{3} \mu_k = \mu_{Cost} + \mu_{Transit-time} + \mu_{CO2} \\ \mu_{Cost} \le \frac{Z_1^U - Z_{Cost}}{Z_1^U - Z_1^*} \\ \mu_{Transit-time} \le \frac{Z_2^U - Z_{Transit-time}}{Z_2^U - Z_2^*} \\ \mu_{CO2} \le \frac{Z_3^U - Z_{CO2}}{Z_3^U - Z_3^*} \\ \textit{Constraint set } (4) - (43) \\ \mu_{Cost} \ge \overline{\theta_1},\ \mu_{Transit-time} \ge \overline{\theta_2} \textit{ and } \mu_{CO2} \ge \overline{\theta_3} \\ 0 \le \mu_{Cost},\ \mu_{Transit-time},\ \mu_{CO2} \le 1 \end{cases} \tag{54}$$

4.2.3. A weighted additive fuzzy goal programming approach

One of the most important issues in fuzzy goal programming approaches is reflection of the importance weights of the fuzzy goals since they may generally be different for the decision analyst. For this reason, Tiwari et al. (1987) added weight coefficients of the goals into the objective function in order to consider this relative importance. This approach aggregates the weighted membership functions of fuzzy goals whose importance degrees may have distinct values. In detail, a single weighted linear utility function is used by this approach to explain the overall preference of the decision analyst for emerging the relative importance of these goals. This utility function is computed by multiplying the membership function of each fuzzy goal with their associated weights. Then, the obtained results are added together (Amid et al., 2009). After construction of the membership functions of the fuzzy goals (see in Fig. 19), the weight value should be assigned to each objective according to its priority. In order to asses these importance weights correctly, we used the desirable achievement degrees which are derived in the previous section under fuzzy group decision making environment. Because, the objectives with

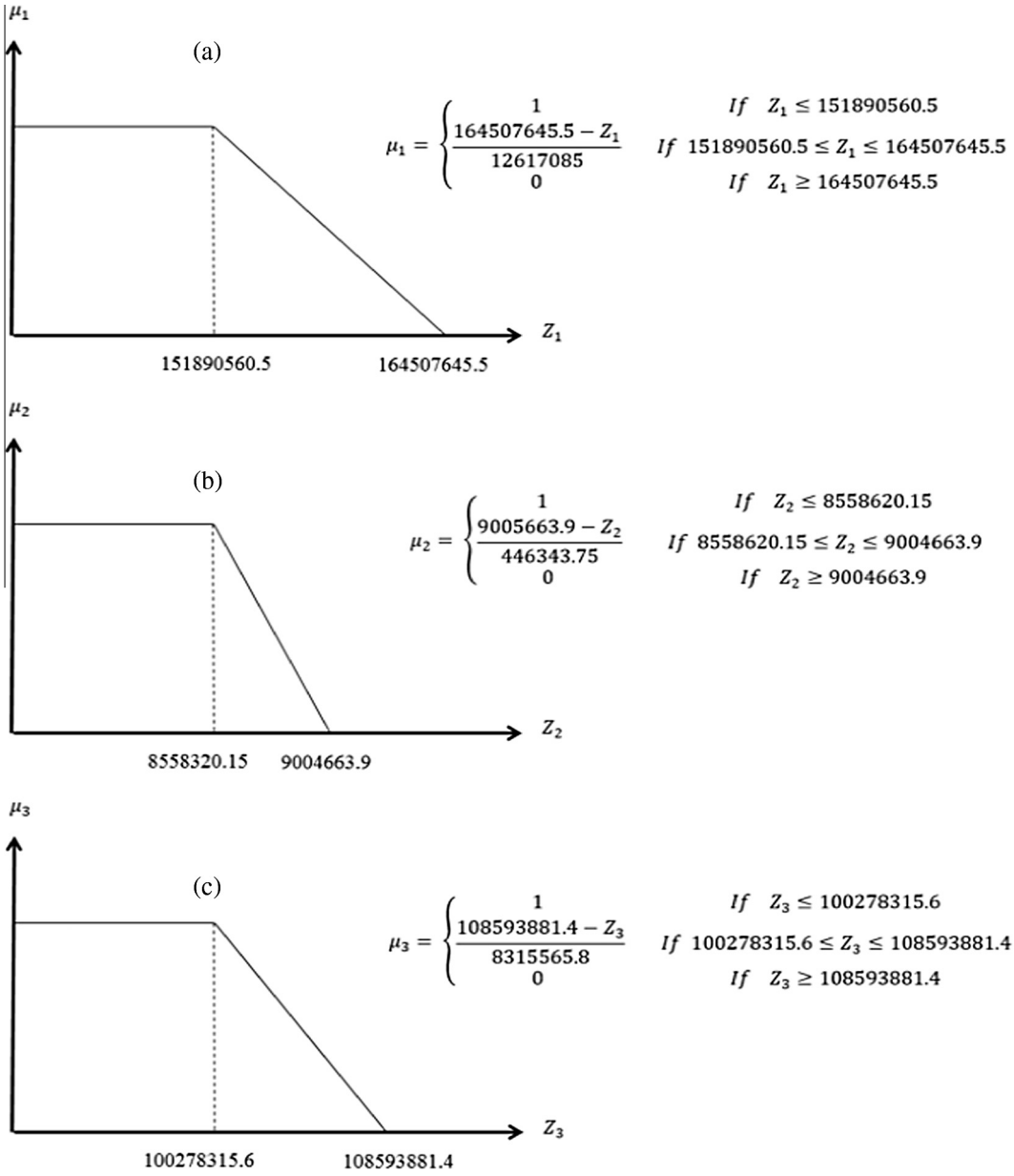


Fig. 19. Membership functions of fuzzy goals: (a) Total transport cost, (b) Total transit time, and (c) Total CO_2 emission.

higher priorities should have also higher weights. This is because desirable achievement degrees represent importance degree of each objective to one another. Then, we implemented a standardization procedure on these values to satisfy that summation of these weights must be equal to 1.0. In order to obtain the final weights, this procedure is formulated in Eq. (55):

$$w_k = \overline{\theta_k} \Big/ \sum_{k=1}^{3} \overline{\theta_k} \tag{55}$$

After this standardization process, the final weight values are provided as 0.425, 0.334 and 0.241 for the total transport cost, total transit time and total CO_2 emission objectives, respectively. In summary, the weighted additive model for the intermodal freight transportation planning problem can be expressed by the following Eq. (56).

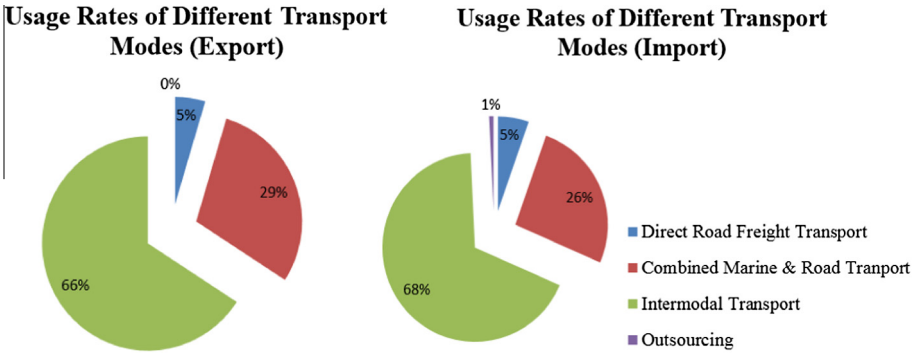


Fig. 20. Usage rates of different transportation modes and outsourcing via FGP approach with DIP.

$$\begin{cases} Maximize \sum_{k=1}^{3} w_k \cdot \mu_k = w_1 \cdot \mu_{Cost} + w_2 \cdot \mu_{Transit-time} + w_3 \cdot \mu_{CO2} \\ \mu_{Cost} \leq \frac{Z_1^U - Z_{Cost}}{Z_1^U - Z_1^*} \\ \mu_{Transit-time} \leq \frac{Z_2^U - Z_{Transit-time}}{Z_2^U - Z_2^*} \\ \mu_{CO2} \leq \frac{Z_3^U - Z_{CO2}}{Z_3^U - Z_3^*} \\ \sum_{k=1}^{3} w_k = 1 \; 0 \leq w_k \leq 1 \\ Constraint\ set\ (4) - (43) \\ 0 \leq \mu_{Cost},\ \mu_{Transit-time},\ \mu_{CO2} \leq 1 \end{cases} \tag{56}$$

where w_k is the weighting parameter which represents the relative importance of fuzzy goals. After application of this FGP approach to the proposed multi-objective load planning model, the following load plans given in Fig. 21 can be provided for different transportation modes and outsourcing option. The computational results are presented in Table 14 for the present multi-objective multimodal load planning problem by applying different MOO approaches. It is clear that weighted additive FGP approach generates similar results with FGP-DIP approach in terms of the cost and time objectives. Indeed, more environmentally friendly load plans can be generated by using the weighted additive FGP over the other FGP approaches.

4.2.4. Interactive fuzzy goal programming approach

The interactive fuzzy goal programming (IFGP) approach was first introduced by El-Wahed and Lee (2006) and applied to multi-objective transportation problems in order to determine the preferred compromise solution. In this approach, three most widely used methods namely interactive programming, goal programming and fuzzy programming were integrated in order to develop a more efficient technique which reflects the advantages of all these methods. In fact, this approach may improve the flexibility and robustness of the decision making process. From the perspective of a decision analyst, the most important advantage of this approach is the ability to control the search direction during the solution phase. In detail, both of the upper bounds and aspiration levels of the goals are updated for producing other efficient solutions

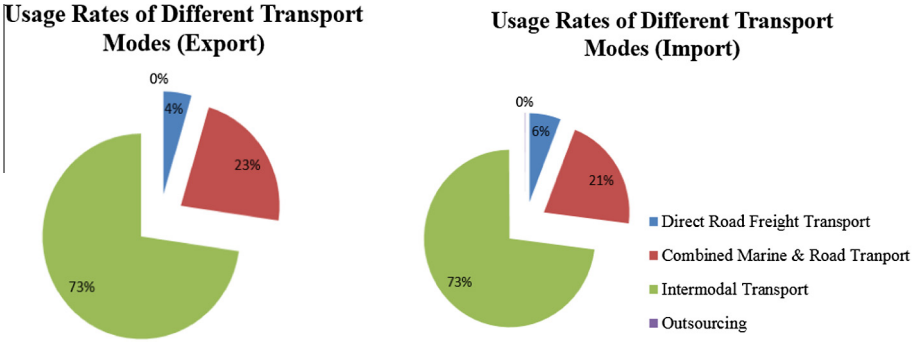


Fig. 21. Usage rates of different transportation modes and outsourcing via weighted additive FGP approach.

Table 14
Fuzzy multi-objective optimization results via different MOO approaches.

Objectives	FGP approach with DIP	Weighted additive FGP approach	Interactive FGP approach	Compromise programming technique		
				$(p = 1)$	$(p = 2)$	$(p = \infty)$
Total transport cost	154035453.9	154641905.4	155479066.42	153463526.9	153236616.8	153132940.9
Total transit times	8714199.96	8718974.96	8685395.9	8902431.08	8904681.96	8901790.58
Total CO_2 emissions	104381954	102117156.2	102642999.41	101122368.78	101388463.1	101520707.6
Satisfaction degrees	0.83; 0.653; 0.506	0.782; 0.642; 0.778	N.A	N.A	N.A	N.A
Model class	MILP	MILP	MILP	MILP	MINLP	MILP
Total number of variables	35,768	35,768	35,772	35,771	35,765	35,775
Total number of integer variables	35,592	35,592	35,592	35,592	35,592	35,595
Total number of constraints	10,771	10,771	10,776	10,765	10,762	107,772
Total non-zero elements	315,822	315,822	315,831	315,822	315,810	315,841
Total number of iterations	1641,506	56,406	92,546	76,618	1072,699	124,796
CPU time (sec.)	838.67	54.48	84.50	70.92	30885.16	104.42

(Selim and Ozkarahan, 2008). The solution obtained from the last iteration is accepted by the decision analyst as the preferred compromise solution and this solution is also perceived as a more realistic one (Subulan et al., 2015b). Interactive FGP approach is able to produce superior results under uncertain environments because of its interactive search strategy. It also incorporates the decision maker into all of the decision making steps for highly reflection of the managerial decisions. In contrast to the FGP-DIP approach which is discussed in Section 4.2.2, it is assumed that there is a single decision maker, i.e., board chairman of the logistics company, while applying the interactive FGP approach. After construction of the membership functions and determination of the aspiration levels, fuzzy MOO model can be transformed into the crisp equivalent form as formulated in Eq. (57) for the first iteration. In Eq. (57), β presents an auxiliary variable. The outline of classical interactive FGP approach can be shown by Fig. 22.

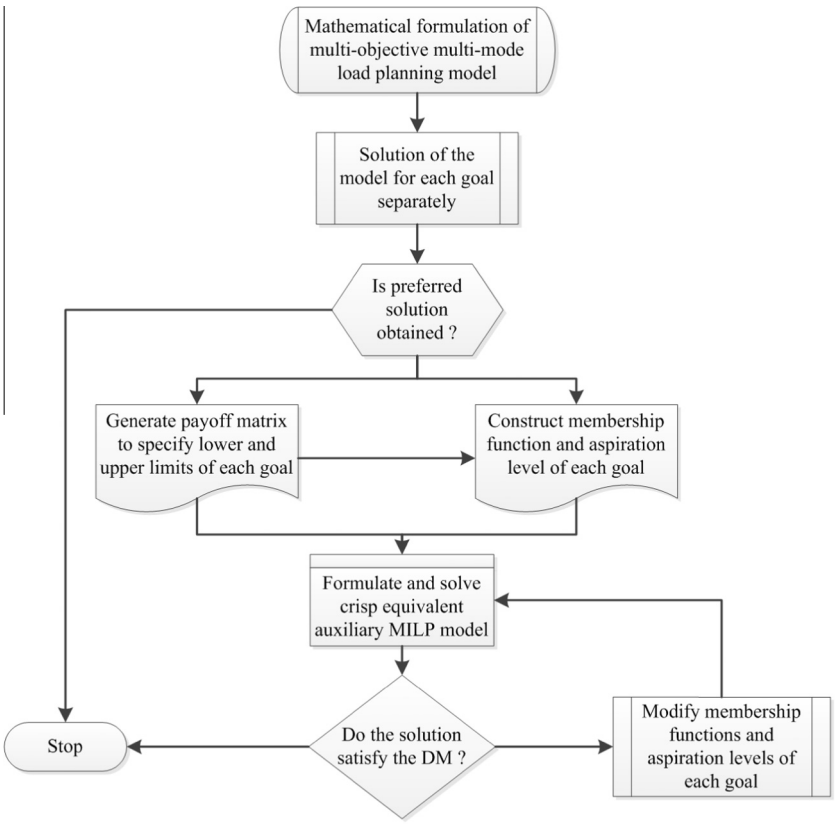


Fig. 22. The application procedure of interactive fuzzy goal programming approach.

$$\begin{cases} \textit{Maximize } \beta \\ Z_{Cost} + \left(Z_1^U - Z_1^*\right) \cdot \beta \le Z_1^U \\ Z_{Transit-time} + \left(Z_2^U - Z_2^*\right) \cdot \beta \le Z_2^U \\ Z_{CO2} + \left(Z_3^U - Z_3^*\right) \cdot \beta \le Z_3^U \\ Z_{Cost} - Z_1^+ + Z_1^- = Z_1^U \\ Z_{Transit-time} - Z_2^+ + Z_2^- = Z_2^U \\ Z_{CO2} - Z_3^+ + Z_3^- = Z_3^U \\ \textit{Constraint set } (4) - (43) \\ 0 \le \beta \le 1 \textit{ and } Z_1^+, Z_1^-, Z_2^+, Z_2^-, Z_3^+, Z_3^- \ge 0 \end{cases} \tag{57}$$

It is assumed that the results of the first iteration whose membership functions are given in Fig. 19 have not satisfied the board chairman of the logistics company yet. Therefore, the former goals' bounds should be replaced by the newly obtained objective values from the first iteration. As a result, the membership functions can be revised in the second iteration as demonstrated by Fig. 23. In other words, the membership functions are re-constructed in the second iteration. According to Fig. 23, new aspiration levels of the objective functions are generated as 155486800.95, 8685542.15 and 102648540.6,

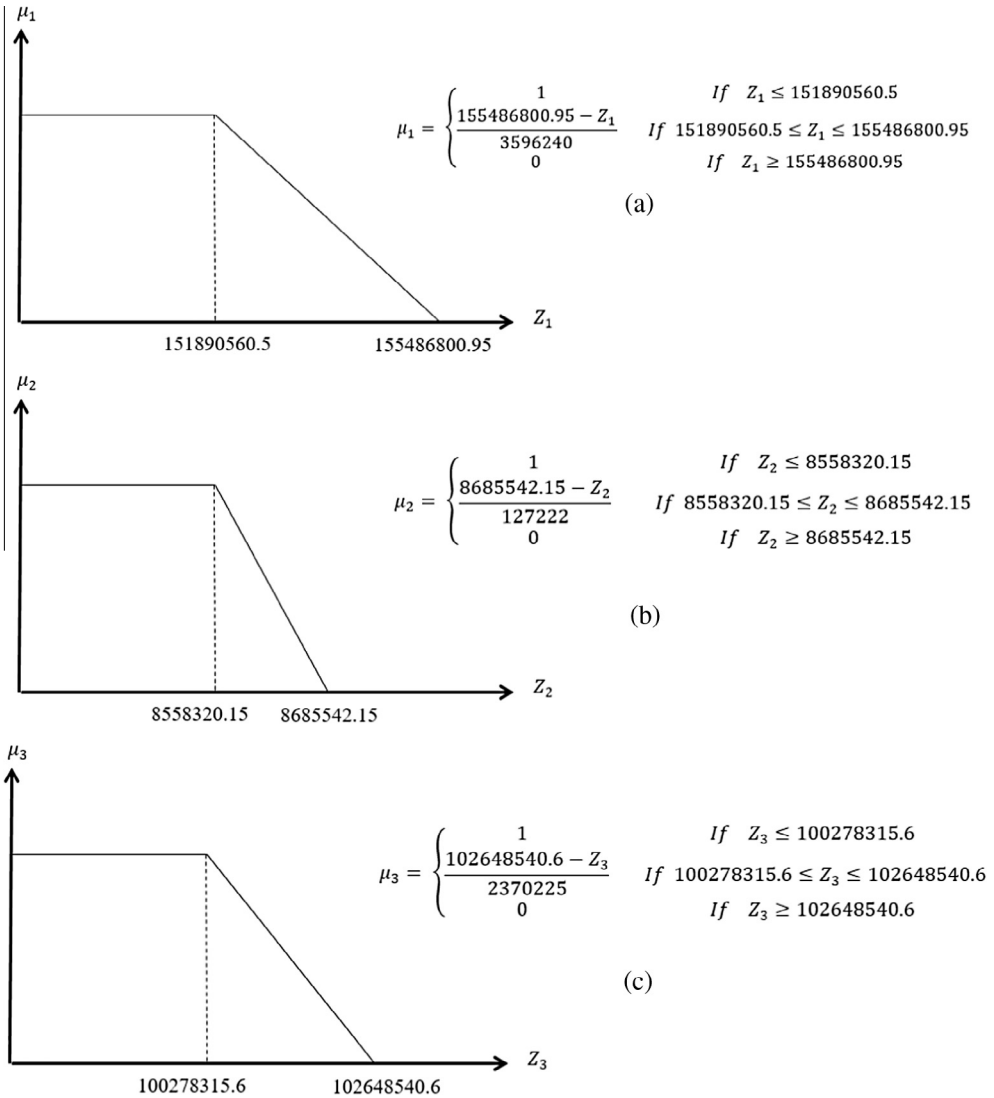


Fig. 23. Membership functions in the second iteration: (a) Total cost, (b) Total transit time, and (c) Total CO_2 emission.

respectively. Depending upon these bound modifications, the crisp equivalent auxiliary model in Eq. (57) is updated and resolved to produce the latter efficient solutions in the second iteration. Optimization results are provided in Table 15 by solving these auxiliary models iteratively.

It is supposed that the board chairman of the company is satisfied at the end of the third iteration with the overall transport cost of €155479066.42 and accepts this solution as the preferred compromise solution. Thereby, the algorithmic procedure configured in Fig. 22 is terminated. More satisfactory solutions can also be provided by making use of the interactive FGP approach when the iteration number is increased within the iterative process. As shown by Figs. 21 and 24, similar load plans are derived from the weighted additive and interactive FGP approaches. In detail, these approaches may provide nearby environmental impact values. In addition, the load plans with shorter transit times can be obtained by applying the interactive FGP approach. On the other hand, this approach leads to more costly solutions than all over the used MOO approaches. In addition to the usage rates of different transportation modes, variations of the overall transport cost, transit time and CO_2 emission values are also presented in Figs. 25–27 by applying different MOO approaches.

When the computational results are evaluated, it is obvious that CP approach with Chebyshev distance metric gives the best cost efficient solution to the decision analyst under crisp environment. Moreover, CP approach with Pythagorean distance metric may also produce more environmentally conscious load plans. However, CP based approaches presented inferior results in terms of the transit time objective when compared to the fuzzy goal programming approaches. Furthermore, it should be mentioned here that significant decreases in total transit time can be yielded via interactive FGP approach. The main components of the total transport cost, transit time and CO_2 emission objectives are shown by Figs. 28–30. According to Fig. 28, the marine transport cost constitutes the large portion of the total annual transportation costs. In addition, all of the used MOO approaches gave the similar results for the marine transport cost. These results are also very close to the ideal solution of the cost minimization objective. It should be noted here that application of different MOO approaches provided almost zero cost of outsourcing in opposition to the single objective deterministic optimization results. As it is clearly seen from Fig. 29 that direct road freight transport yields the minimal transit times for the freight transportation between Turkey and Europe. Conversely, when the maritime and railway transport operations are included, the total transit time will increase due to the long travel distances and increases in awaiting times at the terminals. Since the road freight transport is generally utilized by the outsourcing option, this leads to shorter transit times. Briefly, the combined marine-road freight transport and intermodal transportation will cause longer transit times due to the slow average speed of electrical trains and long travel times of Ro-Ro vessels.

In order to obtain shorter transit times in marine and railway transports, one can utilize FGP-DIP and weighted additive FGP approaches. When the environmental impact components are examined, it is shown by Fig. 30 that marine and road freight transports have significant effect on the environment since the large amounts of CO_2 emission. All of the used MOO approaches gave similar emission values for the marine and railway transportation services. Fortunately, the minimal environmental impact values of the road freight transportation can be obtained from CP approach with Pythagorean distance metric. The import/export load quantities in marine transportation by the company owned Ro-Ro vessels and the required number of block train services are presented in Tables 16 and 17, respectively. Actually, these tables present the load plans in marine and railway transportation services. These results are provided by employing FGP-DIP approach for better reflection of the objective evaluations by the multiple decision makers under uncertain environments. According to the computational experiments, most of the loads are transferred among Istanbul and Trieste seaports in marine transportation by the company-owned Ro-Ro vessels. For this reason, utilization rate of the regarding Ro-Ro vessel is comparatively high than the other Ro-Ro vessels on the marine links (see in Fig. 31). It is also clearly seen from Table 13 that some of the transportation units are loaded or unloaded at the transshipment seaport, i.e., Lavrion, while realizing the maritime logistics operations among the main origin/destination seaports.

On the other hand, the utilization rate of the company-owned Ro-Ro vessel operated between Izmir - Sete seaports is slightly low because of the high transport costs of this speedy vessel. But, this Ro-Ro vessel has lower travel time and environmental impact since its higher average velocity and lower load capacity. Apart from the company-owned Ro-Ro vessels, the majority loads are also transferred by the other logistics providers' Ro-Ro vessels between the seaports of Toulon – Istanbul and Izmir - Trieste. Although, there are many export load transfers available between Izmir - Trieste seaports, the amount of import load transfers between these seaports is considerably low. When the frequency of block train services is investigated, it is obvious that more block trains should be chartered between Trieste-Cologne and Trieste- Ludwigshafen railway stations for both of the import and export transportation operations. Additionally, Table 17 shows that there is a balance between the import and export block train services at each month. This situation fulfills the fixed scheduling limitations of the block train services. In addition to the block train services which have full-than-trainload characteristic (chartering

Table 15
Fuzzy optimization results by all iterations of interactive FGP approach.

Iteration	Auxiliary variable β	Total transport cost	Total transit time	Total CO_2 emission	Solving time (sec.)
1	0.7149	155486800.95	8685542.15	102648540.6	30.58
2	0.00172	155480611.85	8685450	102644128.21	104.33
3	0.000425	155479066.42	8685395.9	102642999.41	84.50
4	No feasible solution				

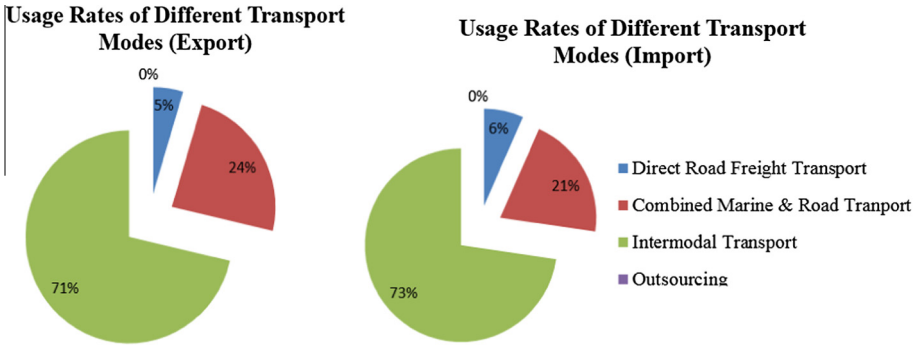


Fig. 24. Usage rates of different transportation modes via interactive FGP approach (third iteration).

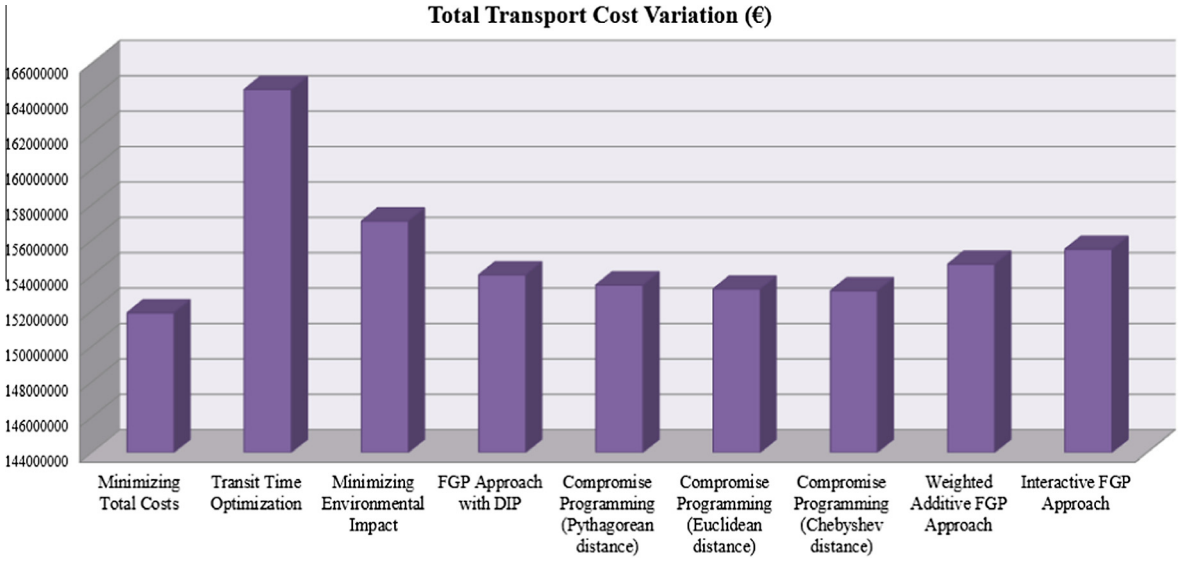


Fig. 25. The variation of total transport cost by single objective optimization and different MOO approaches.

the whole capacity of a block train), some of the loads should be shipped by the public trains between Trieste - Munich railway stations. Actually, these trains have less-than-trainload property and flexible time schedules. In other words, loads of the different logistics companies can be shipped by this kind of trains simultaneously. The public train services also do not require to charter the whole train capacity by the logistics company. Moreover, some of the loads arrived to Cologne transshipment railway station should be transferred to Gothenburg station by the public trains. It is obvious that all of the public train services are performed between Cologne-Gothenborg railway stations instead of using Malmö station.

Finally, the monthly variations in overall transit times and CO_2 emissions are presented in Fig. 32. The company experts evaluated the provided results and enounced that effective and efficient load plans can be obtained by making use of the proposed multi-objective mathematical programming model. Therefore, the logistics company may provide sustainable competitive advantage by improving its load planning mechanism which may contribute to the higher level of customer satisfaction, lower cost and environmental impact values.

4.3. Scenario analysis for managerial insights

In order to investigate the sensitivity of the model parameters on the cost, transit time and environmental objectives and gain some managerial insights related to the intermodal transportation system, the scenario analysis is performed. In fact, the proposed load planning model is resolved with different values of marine transport cost between Izmir – Sete seaports, marine transport cost by the other logistics service providers, outsourcing cost per transportation unit, maximum number of marine cruises by the company-owned Ro-Ro vessels, upper bounds on the number of block train services, lower bounds on the utilization rates of the company-owned Ro-Ro vessels and specific portion of direct road freight transportation in outsourcing option, respectively. These analysis is carried out in seven different scenarios by using the application data given

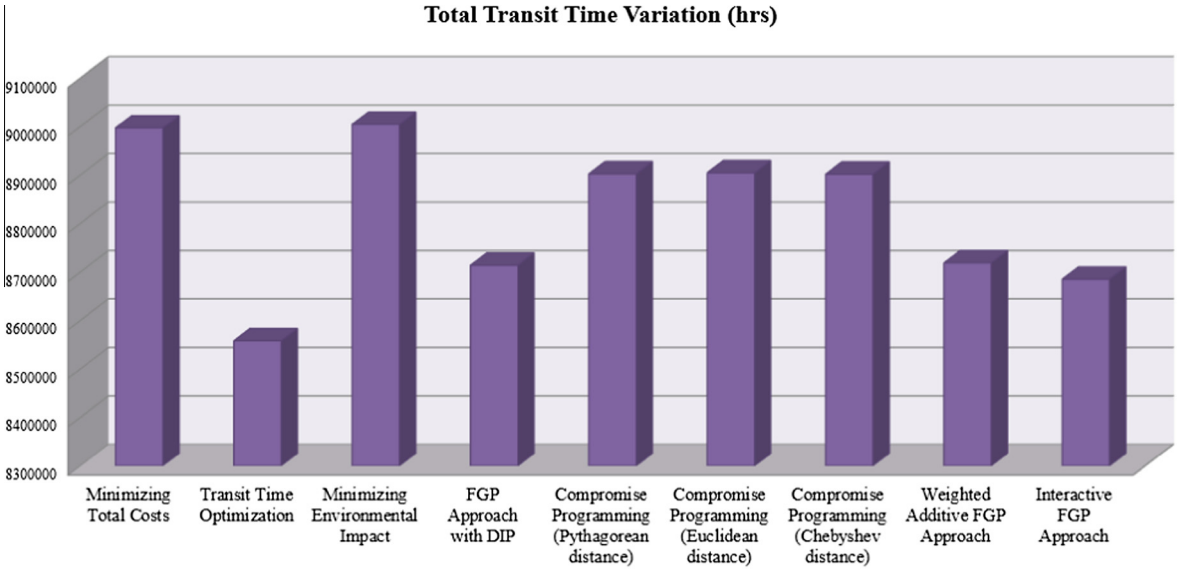


Fig. 26. The variation of total transit time by single objective optimization and different MOO approaches.

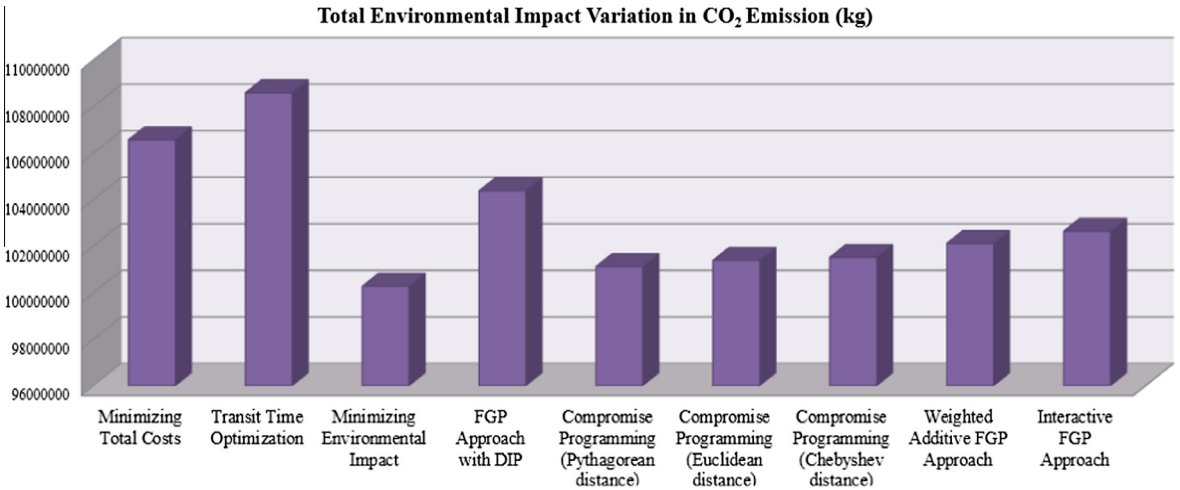


Fig. 27. The variation of total CO_2 emission by single objective optimization and different MOO approaches.

in Table 18. According to the scenario analysis, decreases in marine transport cost between Izmir – Sete seaports and cost of the other logistics service providers' Ro-Ro vessels may provide possible transportation cost reductions (see in Fig. 33). On the other hand, outsourcing cost has not a considerable effect on the total cost of the intermodal transportation system. Although the marine transport cost between Izmir-Sete seaports is decreased down to the 50% of its current value, there will be only 1.8% improvements on the total transportation cost.

In addition, the marine transport cost of the other logistics service providers' Ro-Ro vessels is more influential on the overall transport cost than the cost of operating the company-owned Ro-Ro vessels. In detail, it is possible to provide approximately 3.5% cost reduction when the value of marine transport cost by the other logistics service providers is decreased down to the 10% of its present value. For this reason, the logistics company should focus on the cost of agreements and contracts with the other logistics providers rather than improving its own maritime logistics operations. Moreover, all of these cost oriented scenarios have no impact on the total transit time and environmental objectives as it is clearly seen from Fig. 34. Furthermore, effects of the maximum number of available marine cruises and composition of the outsourcing option, i.e., direct road freight transport and/or combined marine-road transport, are virtually zero on the cost objective as clearly shown by Fig. 33. When the upper bounds on the block train services are decreased down to 30% of its present value, the total transportation cost value will increase up to €154256932.3. On the contrary, there is no significant impact of increases in the number of block train services on the overall transport cost. The logistics company will be exposed to extra marine transport

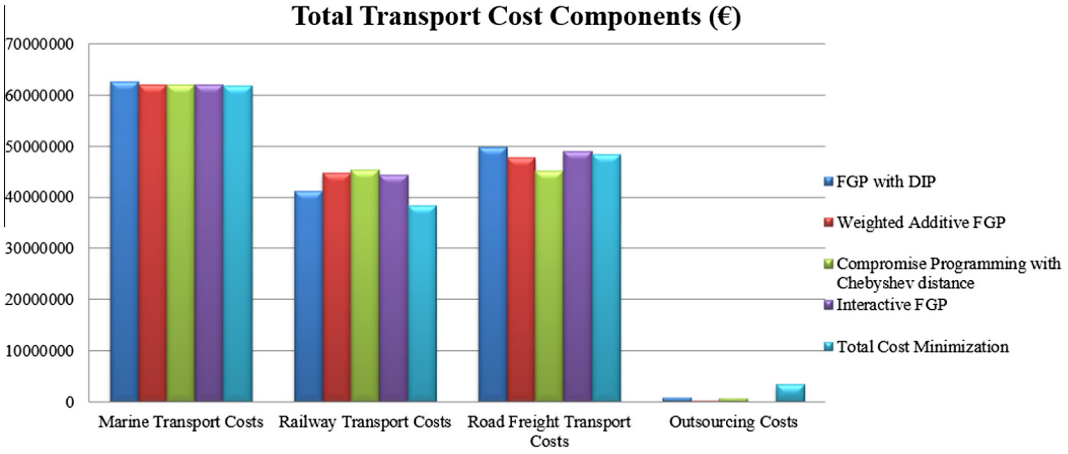


Fig. 28. Components of the total transport cost by cost optimization and different MOO approaches.

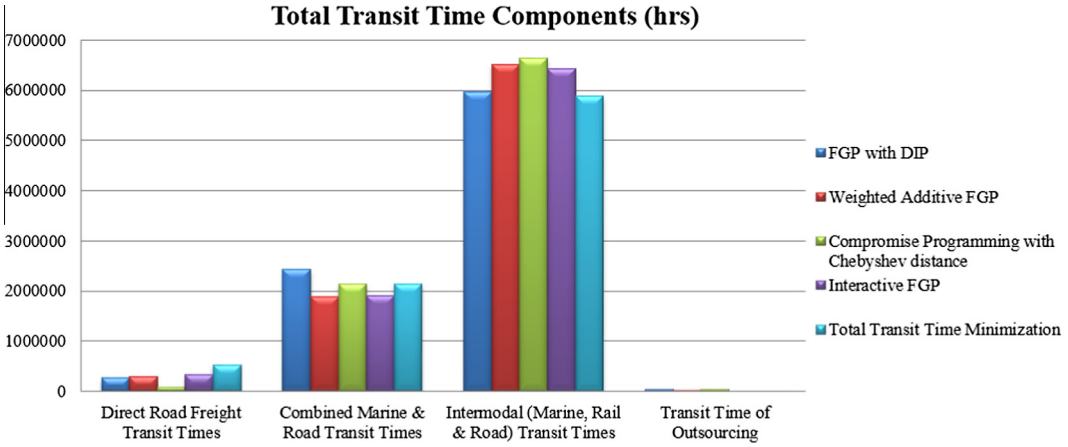


Fig. 29. Components of total transit time by maximizing customer satisfaction and different MOO approaches.

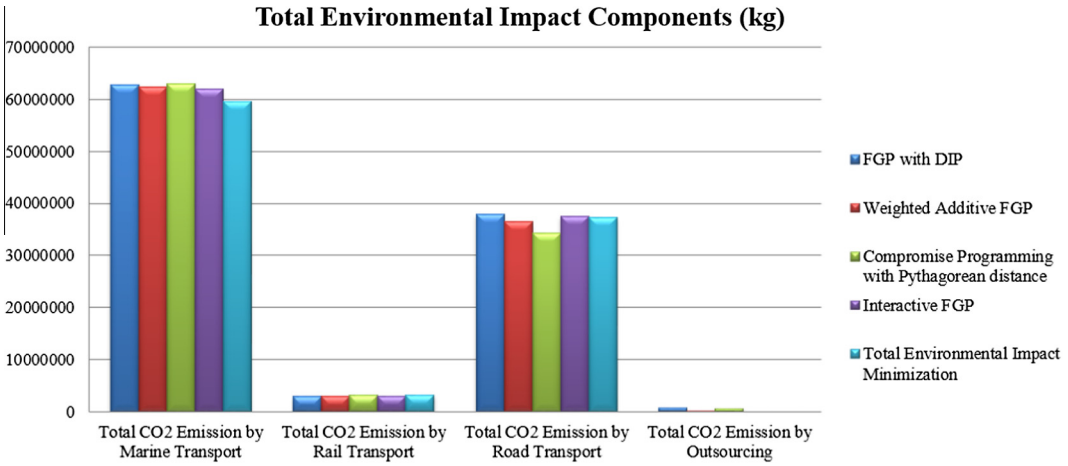


Fig. 30. Components of total environmental impact by CO_2 emission minimization and different MOO approaches.

cost when an upper bound is set to utilization rates of the company-owned Ro-Ro vessel operated between Izmir-Sete seaports. For instance, there will be 3% cost increases when the utilization rate of this Ro-Ro vessel is anticipated to be at least

Table 16
Load planning in railway transportation.

Months	December 15	January 16	February	March	April	May	June	July	August	September	October	November
Number of trips for block train services												
Trieste/Ostrava	12	12	12	13	13	12	11	11	9	13	13	13
Trieste/Ludwig.	27	24	26	26	27	25	27	26	26	27	25	27
Trieste/Cologne	35	35	38	39	38	39	40	39	32	39	39	40
Ostrava/Trieste	13	12	12	13	13	13	12	12	8	13	12	12
Ludwig./Trieste	27	23	26	26	26	26	27	26	26	27	25	27
Cologne/Trieste	34	34	37	39	39	39	41	39	33	40	38	40
The import/export load quantities transferred by public trains												
Trieste/Munich	19	69	26	26	26	35	26	25	21	27	137	68
Cologne/Malmö	0	0	0	0	0	0	0	0	0	0	0	0
Cologne/Guthe.	147	142	164	166	158	156	147	150	135	168	152	156
Munich/Trieste	16	10	28	30	30	30	35	31	22	31	44	29
Malmö/Cologne	0	0	0	0	0	0	0	0	0	0	0	0
Guthe./Cologne	71	68	75	75	83	80	79	74	68	70	61	77

Table 17
Load planning in marine transportation.

Months	December 15	January 16	February	March	April	May	June	July	August	September	October	November
The import/export load quantities in marine transport by the company-owned Ro-Ro vessels												
Istanbul/Lavrion	35	37	36	40	38	40	38	39	17	40	38	39
Istanbul/Trieste	2514	2580	2578	2666	2635	2671	2633	2568	2256	2659	2832	2839
Izmir/Sete	177	275	213	234	225	291	233	197	160	234	300	297
Lavrion/Istanbul	16	20	18	17	17	15	17	16	14	13	18	17
Trieste/Istanbul	2466	2306	2525	2639	2633	2658	2722	2626	2269	2697	2540	2674
Sete/Izmir	187	188	199	207	203	206	201	186	170	187	200	204
The import/export load quantities in marine transport by other logistics service providers' Ro-Ro vessels												
Istanbul/Trieste	0	0	0	0	0	0	0	0	0	0	0	0
Istanbul/Toulon	159	323	172	195	211	215	201	200	156	217	266	235
Izmir/Trieste	341	311	341	337	352	301	354	343	310	348	320	321
Mersin/Trieste	66	75	76	75	75	74	74	69	61	74	77	77
Trieste/Istanbul	0	0	0	0	0	0	0	0	0	0	0	0
Toulon/Istanbul	197	117	195	307	220	207	179	192	261	198	378	198
Trieste/Izmir	31	35	37	39	37	38	40	35	30	36	37	39
Trieste/Mersin	8	9	9	9	9	10	10	9	8	9	9	10

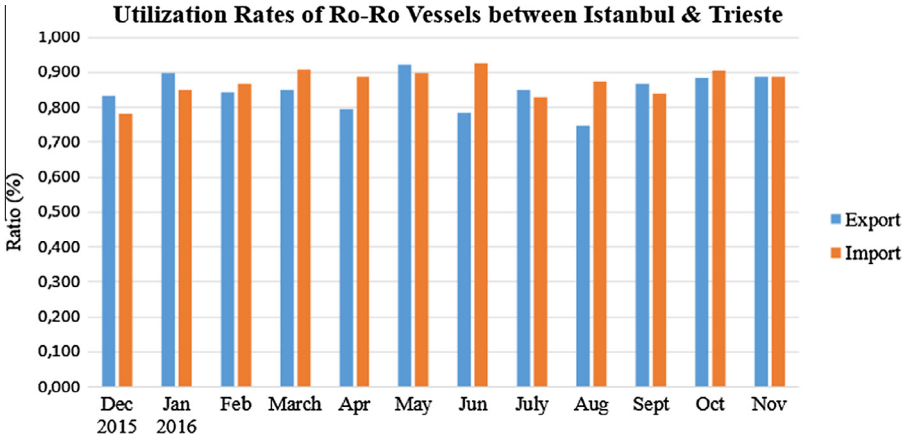


Fig. 31. Utilization rates of the company-owned Ro-Ro vessels operated between Trieste and Istanbul seaports.

70% level. From the perspectives of environmental and transit time objectives, there will be 1.7% and 4.3% increases in total transit time and CO_2 emission values when the upper bounds on the number of block train services are decreased or these services become more infrequent. Fortunately, if the number of block train services is increased, both of the transit time and CO_2 emission objectives will reduce. It should also be mentioned here that there will be maximum 5.1% increases in CO_2 emission value when an upper bound is set to the utilization rate of the Ro-Ro vessel operated between Izmir-Sete seaports.

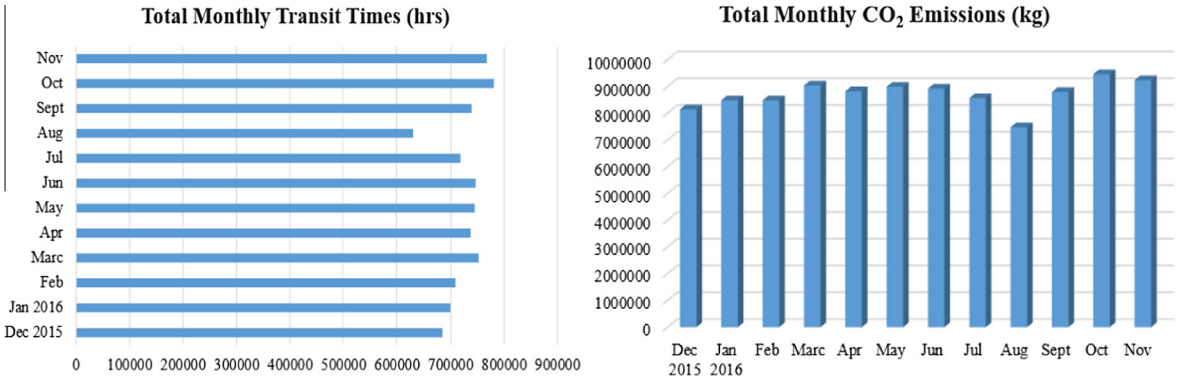


Fig. 32. The variations of the monthly overall transit times and CO_2 emissions, respectively.

Table 18
Application data for scenario analysis.

Scenario	Item	Run-1	Run-2
1	$Co_{ij}^{exp}, Co_{ji}^{imp}$	−50%	+50%
2	$Cp_{ij}^{exp}, Cp_{ji}^{imp}$	−10%	+10%
3	$SC_{ilt}^{exp}, SC_{lit}^{imp}$	−10%	+10%
4	$N_{ijt}^{exp}, N_{jit}^{imp}$	−25%	+25%
5	$UB_{jkt}^{exp}, UB_{kjt}^{imp}$	−30%	+30%
6	$LU_{ijt}^{exp}, LU_{jit}^{imp}$	≫20%	≫50%
7	φ^{Out}	50–50%	25–75%

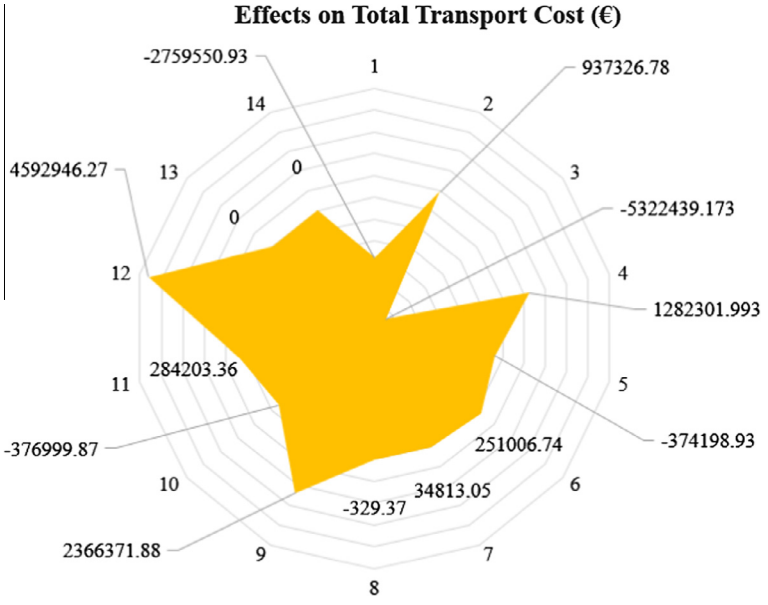


Fig. 33. Increments and reductions in total transport cost as a result of all scenarios.

Furthermore, the utilization rate of the Ro-Ro vessels do not have a considerable effect on the transit time objective. In the original data, it is assumed that 75% of outsourcing operations are performed by direct road freight transport. When this ratio will increase up to 50–75%, there will be no variation in transit time objective because of the unlimited capacity of load units transferred by outsourcing option. However, when only the environmental objective is taken into account, the total CO_2 emission will increase approximately 5% of its optimal value. The corresponding optimal logistics network configurations are presented in Figs. 35–37 for direct road freight transport, outsourcing, combined marine-road transport and intermodal transportation, respectively. Actually, these logistics network configurations are just given for the cost optimization

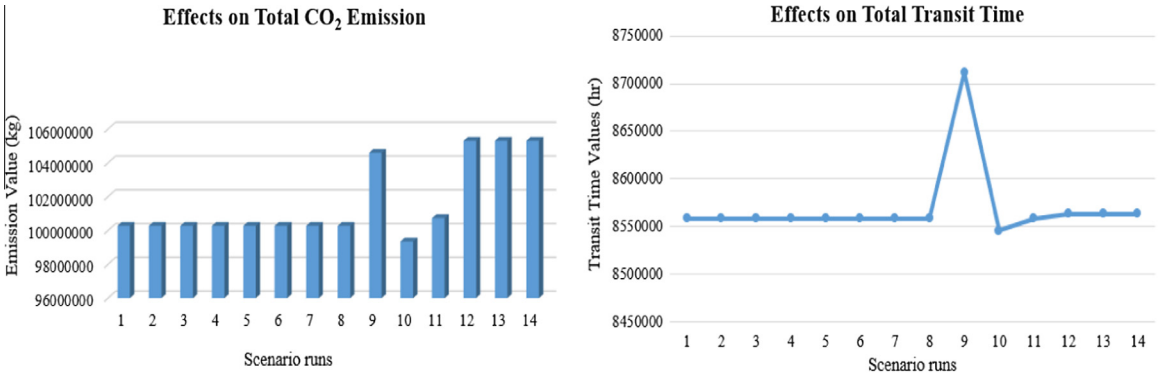


Fig. 34. Variations of total transit time and environmental objectives as a result of all scenarios.

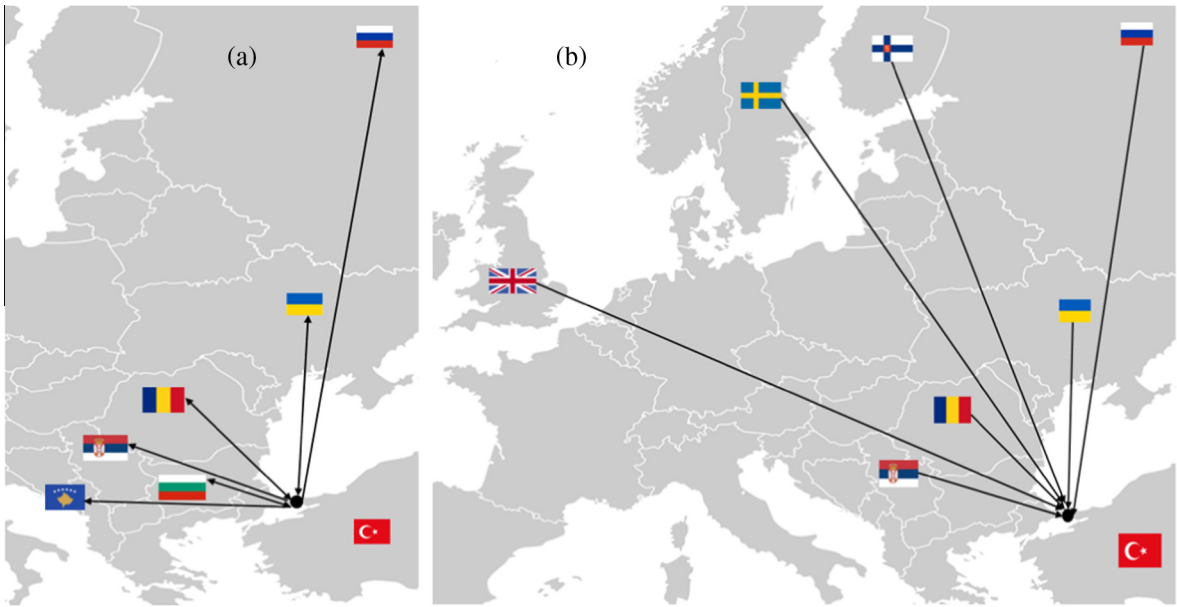


Fig. 35. Logistics network configuration for (a) direct road freight transport and (b) outsourcing option, respectively.

objective due to its higher relative importance. Actually, Fig. 35 represents the import/export transport routes between Turkey and order countries via direct road freight transport and outsourcing alternative. Opposite to the load transfers by outsourcing option, direct road freight transport operations are conducted between Turkey and its neighboring order countries because of the proximity based cost objective. Also, when compared to imported loads by direct road freight transport, the export loads have larger quantities in each month. Due to this fact, there is no any export load transfer by outsourcing option. In fact, only the import transportation operations are performed by means of the outsourcing option for the order countries which are far away from Turkey. Moreover, Fig. 36 displays the combined marine-road freight transport routes in which truckload shipping operations are completed before/after the maritime logistics operations. It is obviously seen from Fig. 36 that most of the maritime logistics operations are performed between Istanbul-Trieste seaports by the company-owned Ro-Ro vessels rather than Sete and Toulon seaports. However, there are also many load transfers available between Izmir-Trieste and Istanbul-Toulon seaports by the other logistics providers' Ro-Ro vessels. The additional trailers or containers which are loaded or unloaded at the transshipment seaport are also depicted in the same figure. These extra loads only satisfy the aggregated transport demands of Greece. There is a balance among the imported and exported load quantities in combined marine-road transport. Lastly, intermodal transportation network of the logistics company can be demonstrated as in Fig. 37. In this figure, Trieste seaport is just reflected on the logistics network as it is the unique seaport with several railway connections. Most of the loads are usually transferred by the block trains between Trieste-Cologne and Trieste-Ludwigshafen railway stations. For export operations, there is no any load transfer between Trieste-Munich railway stations. On the other hand, some loads are transferred by using public trains on this rail line for import operations to Turkey. In addition, it should be emphasized here that after the transshipment Cologne station, all of the loads associated with the freight

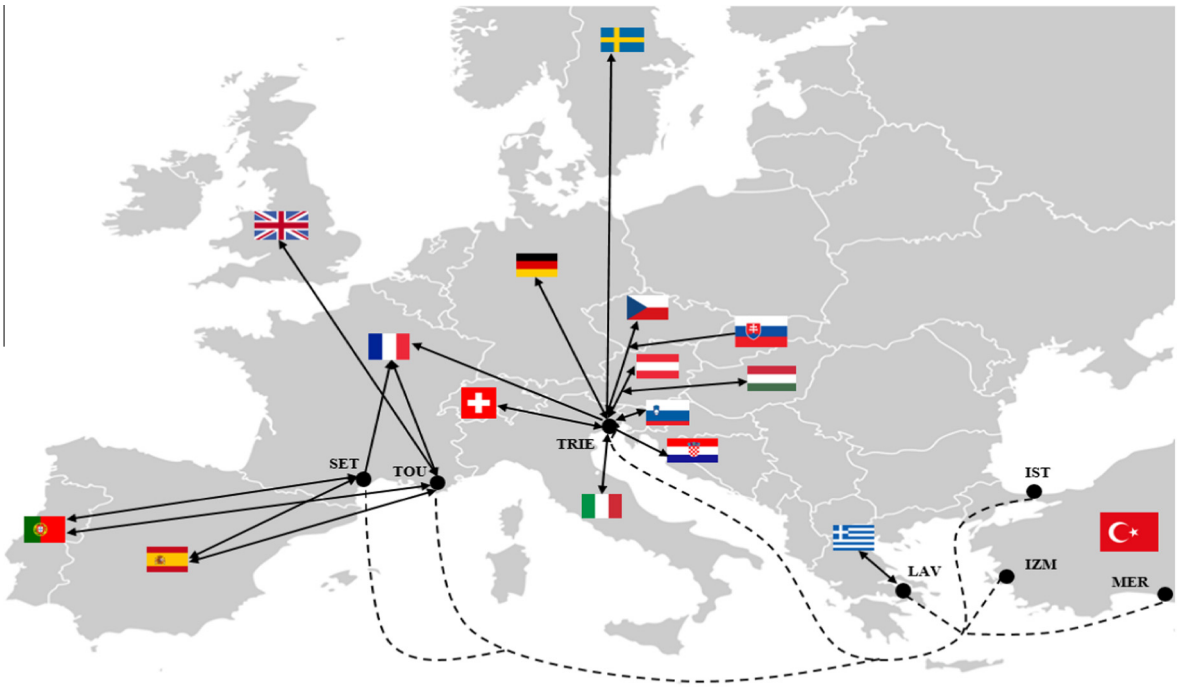


Fig. 36. Import/export transport routes in combined marine and road freight transportation.

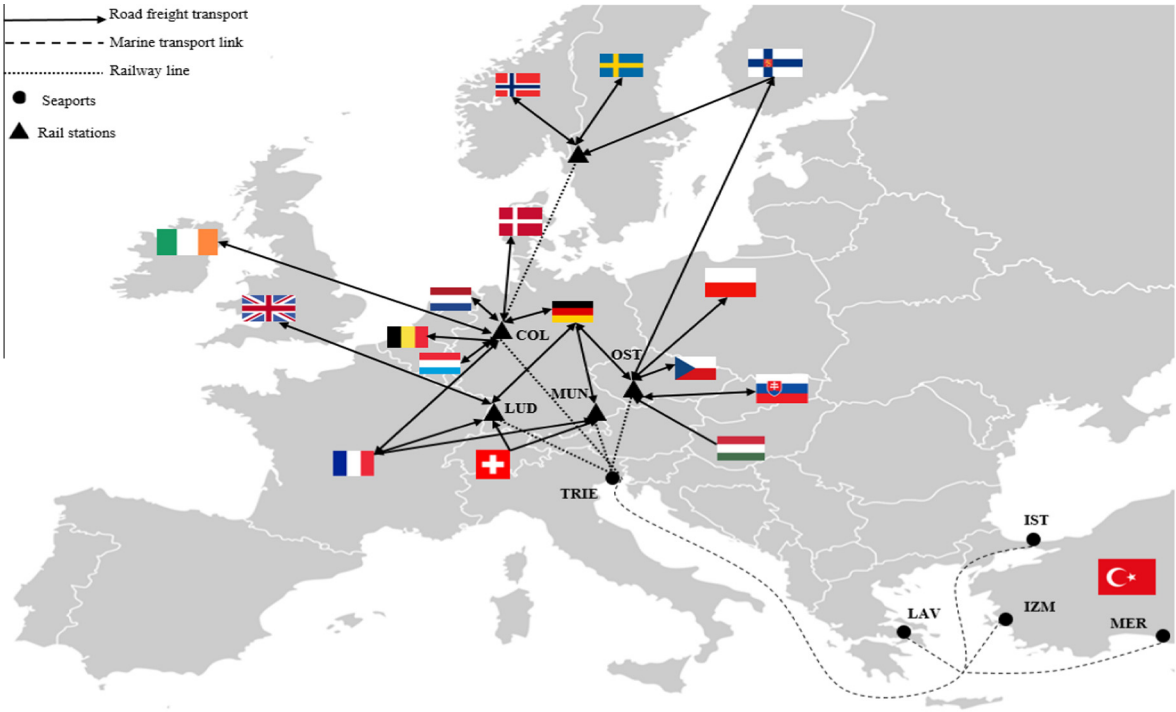


Fig. 37. Intermodal logistics network configuration for the load planning problem in case of cost optimization.

demands of Swedish and Norway will be shipped by public trains via Gothenburg railway station. In other words, there is no need to use Malmö station for satisfying freight demands of these order countries.

5. Concluding remarks

In this research, a novel mixed-integer mathematical programming model is developed to solve a multi-objective, multi-mode and multi-stage periodic load planning or intermodal freight transportation planning problem. Both of the import and export load flows on the intermodal logistics network are also considered by the proposed model. Moreover, the proposed model also incorporates several essential sub-problems such as transportation mode and service type selection, outsourcing, periodic load allocation, consolidation at the transshipment seaports/rail stations, usage of block trains and public trains with fixed or flexible time schedules, etc. In addition to minimize the overall cost of the intermodal transportation system, maximum customer service level satisfaction is also targeted by minimizing the total transit times. Furthermore, amounts of the total CO_2 emissions by different transportation modes are also taken into account so as to provide environmental conscious load plans. In order to generate compromise solutions for these conflicting objectives, several multiple objective optimization approaches are employed under crisp and fuzzy group decision making environments. A real-life application is also presented in a large-sized international logistics company in Turkey for validity and practicality of the proposed model. According to the computational results, intermodal transportation is the most appropriate way of freighting in terms of the economic, customer satisfaction and environmental objectives. Moreover, computational analysis have also suggested that compromise programming approaches with Chebyshev and Pythagorean distance metrics presented superior results for transport cost and environmental goals. Moreover, load plans with shorter transit times can be obtained by using interactive fuzzy goal programming approach under uncertainty of the company's goals. The sensitivity of some model parameters on these goals is also investigated by performing various scenarios. The scenario analysis have indicated that cost of marine transport by the other logistics service providers' Ro-Ro vessels is more influential on the overall transport cost than the cost of operating the company-owned Ro-Ro vessels. Moreover, availability of the monthly block train services has a significant effect on both total transit time and CO_2 emission objectives. Finally, the company experts evaluated the generated results and enounced that effective and efficient load plans can be provided by making use of the proposed model. It is also revealed that the logistics companies may obtain sustainable competitive advantage by improving their load planning mechanisms. In the future, the proposed multi-mode load planning model can be extended to an intermodal fleet planning model by incorporating the fleet management issues such as fleet size/composition, empty truck/trailer repositioning, fleet expansion/reduction and replacement decisions. Furthermore, development of a fuzzy-stochastic planning model which is able to handle uncertainty of the transport demands, transit times, freight capacities and cost parameters may also be scheduled as a future work.

Appendix A

The dataset of the real-life application in an international logistics company can be accessed at the following link: http://web.deu.edu.tr/baykasoglu/dataset_load_planning_model.pdf.

Appendix B

The reader can refer to the following link for the LINGO 15.0 code of the proposed multi-mode load planning model: http://web.deu.edu.tr/baykasoglu/Lingo_code_load_planning_model.pdf.

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